EUROPEAN TELECOMMUNICATION STANDARD

ETS 300 435

February 1996

Source: ETSI TC-RES

Reference: DE/RES-03031

ICS: 30.020, 30.060.50

Key words: DECT, profile, data, LAN

Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Data Services Profile (DSP); Base standard including inter-Working to connectionless networks (service types A and B, Class 1)

ETSI

European Telecommunications Standards Institute

ETSI Secretariat

Postal address: F-06921 Sophia Antipolis CEDEX - FRANCE
Office address: 650 Route des Lucioles - Sophia Antipolis - Valbonne - FRANCE
X.400: c=fr, a=atlas, p=etsi, s=secretariat - **Internet:** secretariat@etsi.fr

Tel.: +33 92 94 42 00 - Fax: +33 93 65 47 16

*

Copyright Notification: No part may be reproduced except as authorized by written permission. The copyright and the foregoing restriction extend to reproduction in all media.

© European Telecommunications Standards Institute 1996. All rights reserved.

Whilst every care has been taken in the preparation and publication of this document, errors in content, typographical or otherwise, may occur. If you have comments concerning its accuracy, please write to "ETSI Editing and Committee Support Dept." at the address shown on the title page.

Contents

Foreword 5

1 Scope 7

2 Normative references 7

3 Definitions and abbreviations 8
3.1 Definitions 8
3.2 Abbreviations 9

4 Description of services 10
4.1 Reference configuration 10
4.2 Service objectives 11

5 PHL layer requirements 11

6 MAC layer requirements 11

7 DLC layer requirements 11

8 NWK layer requirements 11

9 Management entity requirements 12

10 Generic frame relay service inter-Working conventions 12

11 Configuration capabilities 12

Annex A (normative): Implementation of the SIP protected data connectionless downlink service 14

Annex B (normative): InterWorking conventions to specific networks 15

B.1 ISO 8802.3 (Ethernet) 15
B.1.1 Reference configuration 15
B.1.2 Specific inter-Working conventions 15

B.2 ISO 8802.5 (token ring) 16
B.2.1 Reference configuration 16
B.2.2 Specific inter-Working conventions 16

Annex C (normative): Service A: FT Profile ICS Proforma 18

C.1 Standardised symbols for the status column 18

C.2 Capabilities 18
C.2.1 Services 18
C.2.2 Service parameters 19
C.2.3 Protocol parameters 19
C.2.4 Messages 20
C.2.5 Message parameters 23
C.2.6 Functions implemented 24
C.2.7 Timer support 25
C.2.8 Procedure support 26

Annex D (normative): Service A: PT Profile ICS Proforma 28

D.1 Standardised symbols for the status column 28

D.2 Capabilities 28
D.2.1 Services 28
D.2.2 Service parameters 29
D.2.3 Protocol parameters 29
D.2.4 Messages 30
D.2.5 Message parameters 33
D.2.6 Functions implemented 34
D.2.7 Timer support 35
D.2.8 Procedure support 36

Annex E (normative): Service B: FT Profile ICS Proforma 38

E.1 Standardised symbols for the status column 38

E.2 Capabilities 38
E.2.1 Services 38
E.2.2 Service parameters 39
E.2.3 Protocol parameters 39
E.2.4 Messages 40
E.2.5 Message parameters 43
E.2.6 Functions implemented 45
E.2.7 Timer support 46
E.2.8 Procedure support 46

Annex F (normative): Service B: PT Profile ICS Proforma 48

F.1 Standardised symbols for the status column 48

F.2 Capabilities 48
F.2.1 Services 48
F.2.2 Service parameters 49
F.2.3 Protocol parameters 49
F.2.4 Messages 50
F.2.5 Message parameters 53
F.2.6 Functions implemented 55
F.2.7 Timer support 56
F.2.8 Procedure support 56

History 58

Foreword

This draft European Telecommunication Standard (ETS) has been produced by the Radio Equipment and Systems (RES) Technical Committee of the European Telecommunications Standards Institute (ETSI).

Transposition dates	
Date of adoption of this ETS:	15 December 1995
Date of latest announcement of this ETS (doa):	31 May 1996
Date of latest publication of new National Standard or endorsement of this ETS (dop/e):	30 November 1996
Date of withdrawal of any conflicting National Standard (dow):	30 November 1996

Blank page

1 Scope

This ETS defines a profile for Digital European Cordless Telecommunications (DECT) systems conforming to ETS 300 175 [1] to [9]. It is part of a family of profiles that build upon and extend each other, aimed at the general connection of terminals supporting non-voice services to a fixed infra-structure, private and public.

This ETS specifies a generic frame relay service for use within closed user groups. This service is used by other Data Services Profile (DSP) standards when providing inter-working to levels above the Medium Access Control (MAC) layer of the attached network. Annex B contains inter-working conventions to specific attached data networks. This includes inter-working to the MAC layer of international standard connectionless Local Area data Networks (LANs).

This ETS defines both Type A and Type B services. Type A is optimised for low power and simplicity, while Type B is optimised for high speed and throughput. Both are fully compatible and can inter-work with each other.

This ETS defines the requirements on the Physical (PHL), MAC, Data Link Control (DLC) and Network (NWK) layers of DECT.

This ETS also specifies Management Entity (ME) requirements and generic inter-working conventions that ensure the efficient use of the DECT frequency spectrum.

2 Normative references

This ETS incorporates, by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to apply.

[1] ETS 300 175-1: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI); Part 1: Overview".

[2] ETS 300 175-2: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI); Part 2: Physical Layer (PHL)".

[3] ETS 300 175-3: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI); Part 3: Medium Access Control (MAC) layer".

[4] ETS 300 175-4: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI); Part 4: Data Link Control (DLC) layer".

[5] ETS 300 175-5: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI) Part 5: Network (NWK) layer".

[6] ETS 300 175-6: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI) Part 6: Identities and addressing".

[7] ETS 300 175-7: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI) ;Part 7: Security features".

[8] ETS 300 175-8: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI); Part 8: Speech coding and transmission".

[9] ETS 300 175-9: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Common Interface (CI); Part 9: Public Access Profile (PAP)".

[10] TBR 006: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT), General terminal attachment requirements".

[11] I-ETS 300 176: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT), Approval test specification".

[12] CEPT Recommendation T/SGT SF2 (89) 6/0: "Draft Recommendation T/SF Services and Facilities of Digital European Cordless Telecommunications".

[13] ETR 043: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT) Common interface. Services and Facilities requirements specification".

[14] ETR 015: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); Reference document".

[15] ETR 056: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); System description document".

[16] ETR 042: "Radio Equipment and Systems (RES); Digital European Cordless Telecommunications (DECT); A guide to the DECT features that influence the traffic capacity and the maintenance of high radio link transmission quality, including the results of simulations".

[17] ISO 8802: "Information processing - Local Area Networks".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of this ETS, the following definitions apply.

Access Rights Identity (ARI): A globally unique identity that shows the access rights related to a service provider. The ARI consists of an Access Rights Class (ARC) and Access Rights Details (ARD). There are three categories of ARIs:

- Primary ARI (PARI);
- Secondary ARI (SARI);
- Tertiary ARI (TARI).

kbyte: 1 000 bytes.

kbyte: 1 024 bytes.

mobility class 1: Local area applications, for which terminals are pre-registered off-air with one or more specific fixed parts, and establishment of service and user parameters is therefore implicit, according to a profile-defined list.

multiframe: A repeating sequence of 16 successive TDMA frames, that allows low rate or sporadic information to be multiplexed (e.g. basic system information or paging).

service type A: Low speed frame relay, with a net sustainable throughput of up to 24 kbits/s, optimised for burst data, low power consumption and low complexity applications such as hand-portable equipment.

service type B: High performance frame relay, with a net sustainable throughput of up to 552 kbits/s, optimised for high speed and low latency with burst data. Equipment implementation the Type B profile shall inter-operate with Type A equipment.

Time Division Multiple Access (TDMA) frame: A time-division multiplex of 10 ms duration, containing 24 successive full slots. A TDMA frame starts with the first bit period of full slot 0 and ends with the last bit period of full slot 23.

3.2 Abbreviations

For the purposes of this ETS, the following abbreviations apply:

ARC	Access Rights Class
ARD	Access Rights Details
ARI	Access Rights Identity
B_S	slow Broadcast channel
C	higher layer control Channel (see C_S and C_F)
CBC	Connectionless Bearer Control
C_F	higher layer signalling Channel (fast)
C/L	ConnectionLess
CL	higher layer ConnectionLess channel (protected; see CL_S and CL_F)
CL_F	higher layer ConnectionLess channel (fast)
CL_S	higher layer ConnectionLess channel (slow)
C/O	Connection Oriented
C-plane	Control plane
CRC	Cyclic Redundancy Check
CSMA/CA	Carrier Sense Multiple Access/Collision Avoidance
CSMA/CD	Carrier Sense Multiple Access/Collision Detection
C_S	higher layer signalling Channel (slow)
CUG	Closed User Group
DECT	Digital European Cordless Telecommunications
DLC	Data Link Control
DNW	DECT NetWork
E/U-MUX	Switch between E-type and U-type MUltipleXes
FCS	Frame Check Sequence
FDDI	Fibre Distributed Data Interface
FMID	Fixed part MAC IDentity
FP	Fixed Part
FT	Fixed radio Termination
G_F	higher layer information control channel
I	higher layer Information channel (see I_N and I_P)
I_N	higher layer Information channel (unprotected)
I_P	higher layer Information channel (protected)
IPUI	International Portable User Identity
IWF	InterWorking Functions
IWU	InterWorking Unit
LAN	Local Area data Network
LBN	Logical Bearer Number
LCN	Logical Connection Number (DLC/MAC layer)
LLN	Logical Link Number (DLC layer)
LSB	Least Significant Bit
M	MAC control channel
MAC	Medium Access Control
ME	Management Entity
MSB	Most Significant Bit
MUX	time MUltipleXors
N	identities channel
NOS	Network Operating System
OSI	Open Systems Interconnection
P	Paging channel
PARI	Primary Access Rights Identity
PARK	Portable Access Rights Key
PDU	Protocol Data Unit

PHL	PHysical Layer
PMID	Portable part MAC IDentity
PP	Portable Part
PT	Portable radio Termination
Q	system information channel
RFP	Radio Fixed Part
RFPI	Radio Fixed Part Identity
SAP	Service Access Point
SAPI	Service Access Point Identifier
SARI	Secondary Access Rights Identity
SDU	Service Data Unit
SI_N	higher layer connectionless channel (unprotected)
SI_P	higher layer connectionless channel (protected)
TARI	Tertiary Access Rights Identity
T-MUX	Tail MUX
TDD	Time Division Duplex
TDMA	Time Division Multiple Access
TPUI	Temporary Portable User Identity
U-plane	User-plane
WAN	Wide Area Network

4 Description of services

4.1 Reference configuration

The reference configuration for this profile shall be as shown in figure 1.

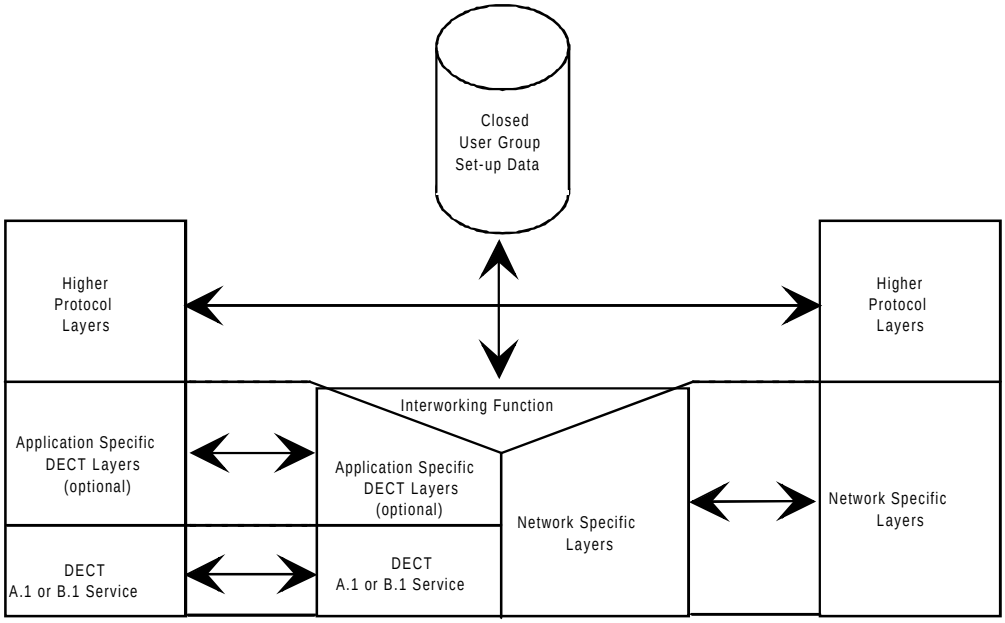

Figure 1: Profile reference configuration showing inter-working to a network via the generic frame relay service

The profile reference configuration is based upon the following principles:

- inter-Working is with an external network via the generic frame relay service (service Type A or B);
- inter-Working with the end system network layer or above is not a requirement of this ETS;
- the set of supported Portable Parts (PPs) shall constitute a closed user group.

4.2 Service objectives

The profile has the service objectives given in tables 1 and 2, as outlined in ETR 043 [13].

Table 1

Offered services	Type A	Type B
Point-to-point Service Data Unit (SDU) transfer PP-FP	Yes	Yes
Point-to-point SDU transfer FP-PP	Yes	Yes
Point-to-multipoint SDU transfer FP-PP	Optional	Optional
Encryption	Optional	Optional

Table 2

Performance	Type A	Type B
Maximum sustainable unidirectional throughput	3 kbytes/s net 24 kbits/s net	69 kbytes/s net 552 kbits/s net
Maximum sustainable full bi-directional throughput	3 kbytes/s net 24 kbits/s net	36 kbytes/s net 288 kbits/s net
Establishment of PT to FT link (average)	< 50 ms average	< 50 ms average
Establishment of FT to PT link (average)	< 160 ms average	< 50 ms average
Undetected bit error ratio	$< 10^{-10}$	$< 10^{-10}$
Uncorrected error rate	$< 10^{-7}$	$< 10^{-7}$
Maximum supported SDU size:	1,5 Kbytes	1,5 Kbytes

5 PHL layer requirements

Full slots shall be used. The Portable radio Termination (PT) shall be capable of operating on any one, and no more than one, physical channel in each time slot. All Radio Fixed Parts (RFPs) shall be capable of operating on at least any one physical channel in each time slot. Use of the Z-field is not defined by the DSP.

6 MAC layer requirements

The minimum instance shall only require the capability to establish and maintain single-bearer connections. The provisions of ETS 300 175-3 [3] shall be implemented with respect to the services, procedures, messages and information elements coding listed in annexes C to F. The provisions of ETS 300 175-6 [6] shall be implemented with respect to the structure and use of identities.

If the Fixed Part (FP) - Portable Part (PP) point-to-multipoint service is implemented the MAC layer shall in addition implement the protected data connectionless downlink service SI_P, as defined in annex A.

7 DLC layer requirements

The DLC U-plane shall be LU2 (Class 1), FU6, (see ETS 300 175-4 [4]). No C-plane functionality is required for the service Types A and B, mobility Class 1.

If the FP-PP point-to-multipoint service is provided the DLC layer shall transmit and receive all FP-PP point-to-multipoint SDUs via LU2 (Class 1) and FU6a framing and segmentation and the PPs shall not send the FU6b acknowledge frames. Such DLC frames shall be transferred via the SI_P service.

8 NWK layer requirements

Inter-operability between units shall be independent of NWK layer functionality. No, DECT NWK layer services are required for service Types A and B, mobility Class 1.

9 Management entity requirements

The LLME shall ensure that a connection is always released, together with all its bearers, after no greater than 5/n seconds, where n = number of bearers, after the last non point-to-multipoint SDU in the IWU buffer has been successfully transferred. The presence of point-to-multipoint SDUs in a buffer shall by itself neither cause the establishment nor the maintenance of a DECT connection-oriented bearer.

In cases where both the PP and the FP are capable of diversity switching, the default operation in the absence of other user intervention shall be for the FP diversity to remain in operation and for the PP to disable its diversity function.

Paging shall always use the default Temporary Portable User Identity (TPUI).

Bi-directional links shall be composed of two unidirectional links in opposite directions on the same MAC connection.

10 Generic frame relay service inter-Working conventions

All data frames shall always be transmitted as DECT DLC layer SDUs, with the earliest bits transferred first in the earliest octet of the earliest U-plane segment.

The DECT equipment shall be capable of supporting SDU frames of at least 1,5 Kbytes. The equipment may optionally support larger SDUs.

Where SDU sizes larger than 1,5 Kbytes are supported by either the FT or the PT the smaller value shall be used in communications, or if this is not possible, the FT equipment shall not allow PT equipment to be registered.

All point-to-multipoint packets shall be transmitted by the FP over the connectionless downlink, and may also be transmitted over previously established connections.

The same International Portable User Identity (IPUI) type shall be used within a given local environment. The IPUI values shall be assigned in such a manner so as to ensure each individual IPUI is unique within all the local environments that support it.

11 Configuration capabilities

In order to assure reliable inter-Working between devices complying with this profile in mobility Class 1, it shall be possible to install the following parameters in the FPs and/or PPs of the system. The values of parameters referring to the PP shall be clearly indicated in the documentation of the PP, and means shall be provided in FPs for such values to be registered. The values of parameters referring to the FP shall be clearly indicated in the documentation of the FP, and means shall be provided in PP for such values to be registered.

Table 3

Variable parameter	Value	Fixed(F)/Portable(P)
Service associated with identity	Type A or B	F, P
Network associated with identity	Selected from annex B	F, P
IPUI	Unique value within local environment	P
ARI	Unique value within local environment	F
Maximum supported SDU size	Number of bytes	F, P
Multi-bearer capability	1-23	F, P
Asymmetric capability	Yes/No	F, P
Diversity capability	Yes/No	F, P
Fast paging available	Yes/No	F, P
Connectionless downlink supported	Yes/No	F, P
Encryption capability	Yes/No	F, P
Static cipher key	64 bits	F, P

A given set of values for the parameters shall be unique for the IPUI with which they are associated.

Annex A (normative): Implementation of the SI_P protected data connectionless downlink service

The SI_P protected data connectionless downlink service is used by the FP-PP point-to-multipoint service to transfer the data frames after the LU2 (Class 1) framing and FU6a segmentation functions have been performed on the point-to-multipoint SDU (see clause 7). The SI_P service is an application of the DECT MAC SI_N service.

The SI_P service shall code each data frame (which shall be 32 octets long) according to the protected B-field multiplex (as defined in ETS 300 175-3 [3], subclause 6.2.1.3) and transmit the coded data as a single SDU via the DECT MAC SI_N channel.

The FP shall only transmit SI_P data starting at the start of a paging cycle. A PP shall understand the presence of SI_P data to be indicated by the coding BA = SI_N and the P_T MAC layer information = Dummy or C/L bearer. The TDMA frame immediately following the frame in which SI_P data was received shall also be monitored to find out whether it contains SI_P data. In this way SI_P data shall be understood to be present in each subsequent TDMA frame until the BA and MAC layer information codings indicate that the SI_P (SI_N) data field is no longer present. No further SI_P information shall then be available until the start of the next paging cycle.

The start of a paging cycle in this context shall be that time-slot in frame 0 of a multiframe that is carrying the start of a paging message. When paging repetition is supported by the fixed part, the number of this multiframe shall be 0 modulo 4.

Annex B (normative): InterWorking conventions to specific networks

B.1 ISO 8802.3 (Ethernet)

The provisions of this clause shall apply if inter-Working to ISO 8802.3 [17] (Ethernet) LANs is provided.

B.1.1 Reference configuration

The reference configuration for this specific inter-Working convention shall be as shown in figure B.1.

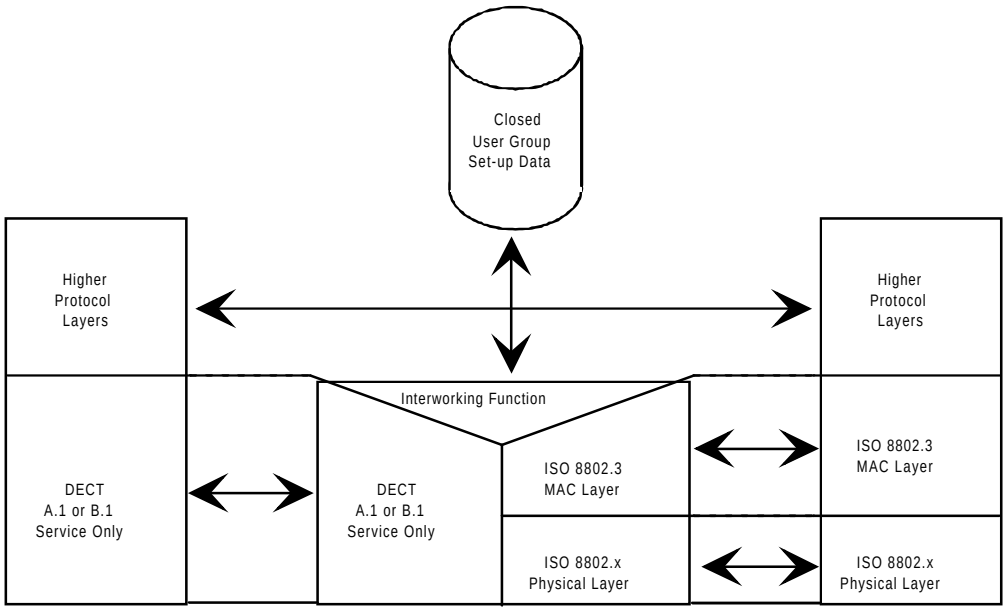

Figure B.1: Profile reference configuration showing inter-Working to ISO 8802.3 type LANs

The profile reference configuration is based upon the principles stated in subclause 4.1 and the following:

- inter-Working is a bridging function with a network conforming to the ISO 8802.3 [17] standard.

B.1.2 Specific inter-Working conventions

The conditions of clause 10 shall be adhered to in addition to the following.

The ISO 8802.3 [17] MAC frame shall be transmitted as a single DECT DLC layer SDU beginning with the ISO 8802.3 [17] MAC Destination Address and ending with the MAC Information field. For MAC frames which are less than 64 bytes in length, the PAD field shall not be transmitted. This mapping is shown in figure B.2.

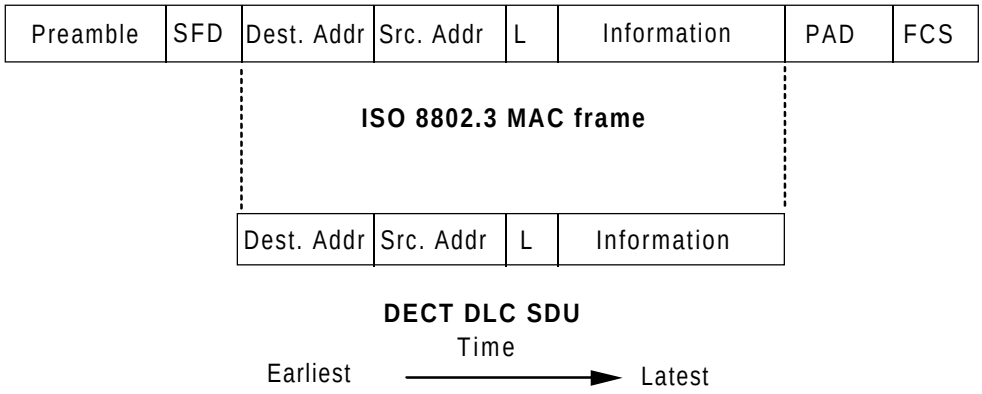

Figure B.2: Mapping of ISO 8802.3 MAC frames into DECT DLC SDU

IPUIs of type O shall be used. The full ISO 8802.3 MAC address shall be mapped into the type O. IPUI with the Least Significant Bit (LSB) of the ISO 8802.3 MAC address corresponding to the LSB of the IPUI. The ISO 8802 MAC source address of frames originating from the PP shall always be checked by the FP at the establishment of each connection, and frames originating from unacceptable sources shall cause the connection to be immediately released.

B.2 ISO 8802.5 (token ring)

The provisions of this clause shall apply if inter-Working to ISO 8802.5 (token ring) LANs is provided.

B.2.1 Reference configuration

The reference configuration for this specific inter-Working convention shall be as shown in figure B.3.

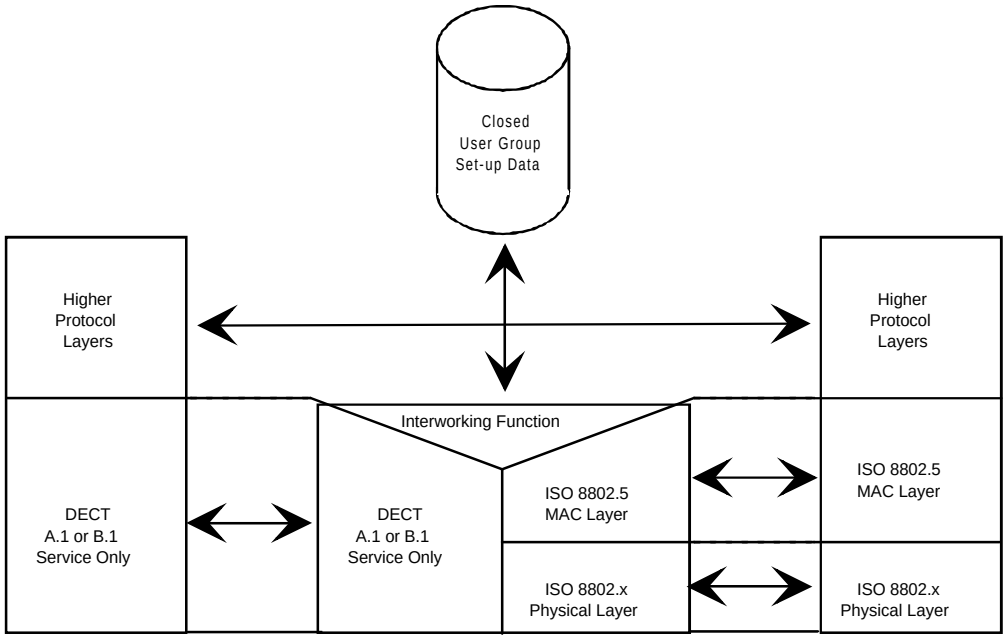

Figure B.3: Profile reference configuration showing inter-Working to ISO 8802.5 type LANs

The profile reference configuration is based upon the principles stated in subclause 4.1 and the following:

- inter-Working is a bridging function with a network conforming to the ISO 8802.5 standard.

B.2.2 Specific inter-Working conventions

All the conditions of clause 10 shall be adhered to in addition to the following:

The ISO 8802.5 MAC frame shall be transmitted as a single DECT DLC layer SDU beginning with the ISO 8802.5 MAC Frame Control (FC) field and ending with the MAC Information field. The FP shall not send the token ring MAC Control frames as identified by the FC byte and it shall not send the Frame Status (FS) byte. It shall be the responsibility of the FP to inter-work these to the token ring network. This mapping is shown in figure B.4.

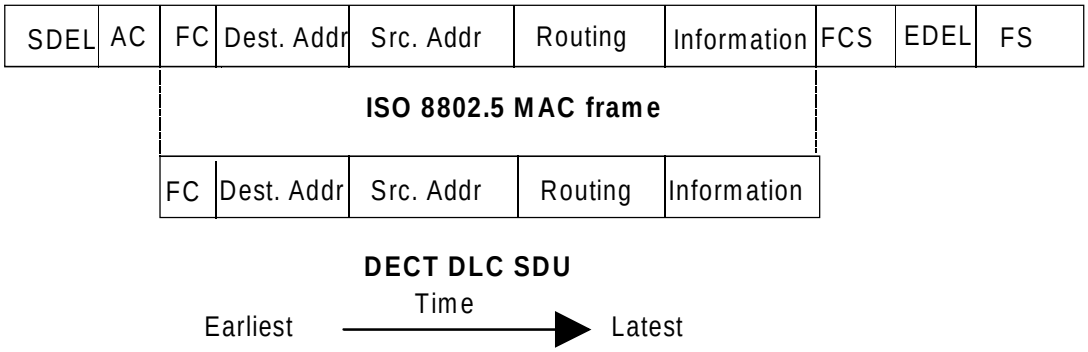

Figure B.4: Mapping of ISO 8802.4 MAC frames into DECT DLC SDU

IPUIs of type O shall be used. The full ISO 8802.5 MAC address shall be mapped into the type O IPUI with the LSB of the ISO 8802.5 MAC address corresponding to the LSB of the IPUI. The ISO 8802 [17] MAC source address of frames originating from the PP shall always be checked by the fixed part at the establishment of each connection, and frames originating from unacceptable sources shall cause the connection to be immediately released.

Annex C (normative): Service A: FT Profile ICS Proforma

Notwithstanding the provisions of the copyright clause related to the text of this ETS, ETSI grants that users of this ETS may freely reproduce the PICS proforma in this annex so that it can be used for its intended purposes and may further publish the completed PICS.

C.1 Standardised symbols for the status column

The standardised symbols for the status column are as follows:

- m or M for mandatory;
- o or O for optional (Boolean);
- x or X for prohibited use;
- n/a, N/A or - (dash) for not applicable;
- cn or Cn for conditional on the implementation of feature n;
- i or I for out of scope (the capability is allowed to be implemented but is not called upon by the profile functionality).

Where reception of a message is marked as "O", this shall be understood to mean that correct understanding and processing of that message is optional.

C1: IF Encryption supported THEN M ELSE I.

C2: IF Connectionless downlink supported THEN M ELSE I.

C3: IF Class 2 mobility supported THEN M ELSE I.

Except when stated otherwise, the reference column refers to the relevant subclause in ETS 300 175-3 [3].

C.2 Capabilities

C.2.1 Services

Q.9 Service support

Service supported				
Item no.	**Name of service**	**Ref.**	**Status**	**Support**
1	Type 1f I_N_minimum delay	5.6.2.1	I	
2	Type 2f I_N_normal delay	5.6.2.1	I	
3	Type 3f I_P_error_detection	5.6.2.1	I	
4	Type 4f I_P_error_correction	5.6.2.1	M	
5	Type 5 I_N_normal delay	5.6.2.2	I	
6	Type 6 I_P_error_detection	5.6.2.2	I	
7	Type 7f I_P_error_correction	5.6.2.2	I	
8	Type 1h I_N_minimum delay	5.6.2.1	I	
9	Type 2h I_N_normal delay	5.6.2.1	I	
10	Type 3h I_P_error_detection	5.6.2.1	I	
11	Type 4h I_P_error_correction	5.6.2.1	I	
12	C_S duplex	5.3.1.1	C3	
13	C_F duplex	5.3.1.1	C3	
14	Downlink CL_S only	5.7.2.1	I	

(continued)

Q.9 Service support (concluded)

Service supported				
Item no.	**Name of service**	**Ref.**	**Status**	**Support**
15	Downlink CL_F + CL_S	5.7.2.1	I	
16	Downlink CL_S + SI_N	5.7.2.1	C2	
17	Uplink CL_F only	5.7.2.2	I	
18	Uplink CL_S only	5.7.2.2	I	
19	Uplink without SDU	5.7.2.2	I	
20	G_F simplex	5.3.1.3	M	
21	Fast paging	11.3.3.1	O	
22	Normal paging	11.3.3.1	M	
23	Low duty cycle paging	11.3.3.1	O	
24	System identities	11.3.2	M	
25	System information	11.3.2	M	
26	Encryption	6.2.3	C1	
27	Fast setup	11.3.3.2	I	

C.2.2 Service parameters

Q.10 Service parameters

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	No. of bearers in T2f service	5.6.2.1	I			
2	No. of bearers in T3f service	5.6.2.1	I			
3	No. of bearers in T4f service	5.6.2.1	M		1	
4	Forward bearer no. in T5 service	5.6.2.2	I			
5	Special bearer no. in T5 service	5.6.2.2	I			
6	Forward bearer no. in T6 service	5.6.2.2	I			
7	Special bearer no. in T6 service	5.6.2.2	I			
8	Forward bearer no. in T7 service	5.6.2.2	I			
9	Special bearer no. in T7 service	5.6.2.2	I			

C.2.3 Protocol parameters

Q.11 Protocol parameters

Protocol parameters supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Value**	
					Allowed	**Supported**
1	N200	10.2	M		10	
2	N201	10.6	I			
3	N202	11.4	M		10	
4	N203	9.2	I			

C.2.4 Messages

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
1	TA code: CT data packet number 0	7.1.2	C3		C3	
2	TA code: CT data packet number 1	7.1.2	C3		C3	
3	TA code: identities information on connectionless bearer	7.1.2	C2		N/A	
4	TA code: identities information	7.1.2	M		M	
5	TA code: multiframe sync. and system information	7.1.2	M		N/A	
6	TA code: main escape	7.1.2	X		I	
7	TA code: MAC layer control	7.1.2	M		M	
8	TA code: paging/first PT transmission	7.1.2	M		M	
9	BA code: U-type, IN, SIN or IP packet number 0	7.1.4	M		M	
10	BA code: U-type, IP error detect or IP packet number 1	7.1.4	M		M	
11	BA code: E-type, all C_F or CL_F, packet number 0	7.1.4	C3		C3	
12	BA code: E-type, all C_F, packet number 1	7.1.4	C3		C3	
13	BA code: E-type, not all C_F or CL_F; C_F packet number 0	7.1.4	C3		C3	
14	BA code: E-type, not all C_F; C_F packet number 1	7.1.4	C3		C3	
15	BA code: E-type, all MAC control	7.1.4	M		M	
16	BA code: no B_field	7.1.4	O		N/A	
17	NT information	7.2.2	M		M	
18	Static system information	7.2.3.2	M		N/A	
19	Extended RF carrier information	7.2.3.3	I		N/A	
20	Fixed part capabilities	7.2.3.4	M		N/A	
21	Extended fixed part capabilities	7.2.3.5	I		N/A	
22	SARI message	7.2.3.6	O		N/A	
23	Multi-frame number	7.2.3.7	C1		N/A	
24	Q-channel escape	7.2.3.8	I		N/A	
25	Zero length page indication	7.2.4.2.3	M		N/A	
26	Short page indication	7.2.4.2.3	M		N/A	
27	Full page indication	7.2.4.2.3	I		N/A	
28	Not the last 36 bits of a long page indication	7.2.4.2.3	I		N/A	
29	The first 36 bits of a long page indication	7.2.4.2.3	I		N/A	
30	The last 36 bits of a long page indication	7.2.4.2.3	I		N/A	
31	All of a long page indication	7.2.4.2.3	I		N/A	
32	PT header extend flag	7.2.4.2.2	M		N/A	
33	PT MAC information: fill	7.2.4.3.2	O		N/A	
34	PT MAC information: blind full slot	7.2.4.3.3	M		N/A	
35	PT MAC information: other bearer	7.2.4.3.4	O		N/A	
36	PT MAC information: recommended other bearer	7.2.4.3.4	O		N/A	
37	PT MAC information: good RFP bearer	7.2.4.3.4	O		N/A	
38	PT MAC information: dummy or C/L bearer position	7.2.4.3.4	M		N/A	
39	PT MAC information: RFP identity	7.2.4.3.5	I		N/A	
40	PT MAC information: dummy or C/L bearer marker	7.2.4.3.7	C2		N/A	
41	PT MAC information: escape	7.2.4.3.6	I		N/A	
42	PT MAC information: bearer handover type 0000	7.2.4.3.8	I		N/A	
43	PT MAC information: bearer handover type 0001	7.2.4.3.8	I		N/A	
44	PT MAC information: bearer handover type 0010	7.2.4.3.8	I		N/A	
45	PT MAC information: bearer handover type 0011	7.2.4.3.8	I		N/A	
46	Basic access request	7.2.5.2	N/A		I	
47	Basic bearer handover request	7.2.5.2	N/A		I	

(continued)

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
48	Basic connection handover request	7.2.5.2	N/A		I	
49	Basic unconfirmed access request	7.2.5.2	N/A		I	
50	Basic bearer confirm	7.2.5.2	I		N/A	
51	Basic wait	7.2.5.2.3	I		I	
52	Basic bearer release	7.2.5.2	I		I	
53	Advanced access request	7.2.5.3.2	I		I	
54	Advanced bearer handover request	7.2.5.3.3	I		I	
55	Advanced connection handover request	7.2.5.3.4	I		I	
56	Advanced bearer confirm	7.2.5.3.6	I		I	
57	Advanced unconfirmed access request	7.2.5.3.5	I		I	
58	Advanced wait	7.2.5.3.7	I		I	
59	Advanced attributes-T request	7.2.5.3.8	I		I	
60	Advanced attributes-T confirm	7.2.5.3.8	I		I	
61	Advanced bandwidth-T request	7.2.5.3.9	I		I	
62	Advanced bandwidth-T confirm	7.2.5.3.9	I		I	
63	Advanced channel list	7.2.5.3.10	I		I	
64	Advanced unconfirmed dummy	7.2.5.3.11	I		I	
65	Advanced unconfirmed handover	7.2.5.3.12	I		I	
66	Advanced bearer release	7.2.5.3.13	I		I	
67	Transmit on the specified slot/frequency test message	7.2.5.4.2	N/A		N/A	
68	Loopback B_field test message	7.2.5.4.3	N/A		M	
69	Defeat antenna diversity test message	7.2.5.4.4	N/A		M	
70	Force bearer handover test message	7.2.5.4.5	N/A		I	
71	Clear test mode test message	7.2.5.4.6	N/A		M	
72	Single bearer antenna switch QC message	7.2.5.5	O		O	
73	All bearer antenna switch QC message	7.2.5.5	O		O	
74	Bearer handover QC message	7.2.5.5	I		I	
75	Connection handover QC message	7.2.5.5	I		I	
76	Single bearer frequency control QC message	7.2.5.5	O		O	
77	All bearer frequency control QC message	7.2.5.5	O		O	
78	C/L single transmission with 1 CL_F segment	7.2.5.6	I		I	
79	C/L single transmission with 2 CL_F segments	7.2.5.6	I		I	
80	C/L single transmission with 3 CL_F segments	7.2.5.6	I		I	
81	C/L single transmission with 4 CL_F segments	7.2.5.6	I		I	
82	1st C/L half slot transmission with 1 CL_F segment	7.2.5.6	I		I	
83	1st C/L full slot transmission with 4 CL_F segment	7.2.5.6	I		I	
84	2nd C/L transmission with 1 CL_F segment	7.2.5.6	I		I	
85	2nd C/L transmission with 2 CL_F segment	7.2.5.6	I		I	
86	2nd C/L transmission with 3 CL_F segment	7.2.5.6	I		I	
87	2nd C/L transmission with 4 CL_F segment	7.2.5.6	I		I	
88	C/L single transmission without CL_F segment	7.2.5.6	I		I	
89	1st C/L transmission without CL_F segment (CL_S service)	7.2.5.6	I		I	
90	Change dummy position	7.2.5.6	O		O	
91	Extended system information	7.2.5.6	O		O	
92	Start encryption: request	7.2.5.7	N/A		C1	

(continued)

Q.12 Messages in A_tail (concluded)

Supported messages						
93	Start encryption: confirm	7.2.5.7	C1		N/A	
94	Start encryption: grant	7.2.5.7	N/A		C1	
95	Stop encryption: reques t	7.2.5.7	N/A		C1	
96	Stop encryption: confirm	7.2.5.7	C1		N/A	
97	Stop encryption: grant	7.2.5.7	N/A		C1	
98	First transmission for B_field setup	7.2.5.8	M		M	
99	M-channel escape	7.2.5.9	I		I	

Q.13 Messages in B_field

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
1	B_field access request	7.3.3.2	I		M	
2	B_field bearer handover request	7.3.3.2	I		I	
3	B_field connection handover request	7.3.3.2	I		I	
4	B_field unconfirmed access request	7.3.3.2	I		I	
5	B_field bearer confirm	7.3.3.3	M		I	
6	B_field wait	7.3.3.4	M		M	
7	B_field attributes-B request	7.3.3.5	I		I	
8	B_field attributes-B confirm	7.3.3.5	I		I	
9	B_field bandwidth-B request	7.3.3.6	I		I	
10	B_field bandwidth-B confirm	7.3.3.6	I		I	
11	B_field channel list	7.3.3.7	I		I	
12	B_field unconfirmed dummy	7.3.3.8	I		I	
13	B_field unconfirmed handover	7.3.3.9	I		I	
14	B_field bearer release	7.3.3.10	M		M	
15	B_field null message	7.3.4	M		M	
16	B_field single bearer antenna switch QC message	7.3.5.2	O		O	
17	B_field all bearer antenna switch QC message	7.3.5.2	O		O	
18	B_field bearer handover QC message	7.3.5.2	I		I	
19	B_field connection handover QC message	7.3.5.2	I		I	
20	B_field single bearer frequency control QC message	7.3.5.2	O		O	
21	B_field all bearer frequency control QC message	7.3.5.2	O		O	
22	B-field reset request	7.3.5.3	M		M	
23	B_field reset confirm	7.3.5.3	M		M	
24	MAC-MOD2-ACK message	7.3.5.4	M		M	
25	B-field request for QT field	7.3.6.2	N/A		O	
26	B-field QT field response	7.3.6.3	O		N/A	
27	B_field TARI message	7.3.6.4	O		N/A	
28	B_field G_F data	7.3.7	M		M	
29	B_field M-channel escape	7.3.8	O		O	

C.2.5 Message parameters

Q.14 Parameters of static system information message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Normal-reverse	7.2.3.2.2	M		0	
2	Slot number	7.2.3.2.3	M		0 - 11	
3	Start position	7.2.3.2.4	M		0	
4	Escape	7.2.3.2.5	M		0 - 1	
5	Number of transceivers	7.2.3.2.6	M		0 - 3	
6	Extended RF carrier flag	7.2.3.2.7	M		0	
7	RF carriers available	7.2.3.2.8	M		1 - 1 023	
8	Spare	7.2.3.2.11	M		0	
9	Carrier number	7.2.3.2.10	M		0 - 9	
10	Spare	7.2.3.2.11	M		0	
11	Primary receiver scan number	7.2.3.2.12	M		0 - 9	

Q.15 Parameters of Fixed Part Capabilities Message

Encryption capability						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Extended FP information	7.2.3.4.2	M		0, 1	
2	Full slot	7.2.3.4.2	M		1	
3	Frequency control	7.2.3.4.2	M		0, 1	
4	Page repetition	7.2.3.4.2	M		0, 1	
5	C/O setup on dummy allowed	7.2.3.4.2	M		C2 = 0	
6	C/L uplink	7.2.3.4.2	M		0, 1	
7	C/L downlink	7.2.3.4.2	M		C2 = 1	
8	Basic A_field setup	7.2.3.4.2	M		0, 1	
9	Advanced A_field setup	7.2.3.4.2	M		0, 1	
10	B_field setup	7.2.3.4.2	M		1	
11	C_F messages	7.2.3.4.2	C3		0, 1	
12	I_N_minimum_delay	7.2.3.4.2	M		0, 1	
13	I_N_normal_delay	7.2.3.4.2	M		0, 1	
14	I_P_error_detection	7.2.3.4.2	M		0, 1	
15	I_P_error_correction	7.2.3.4.2	M		1	
16	Multibearer connection	7.2.3.4.2	M		0	

The higher layer information field of the FP capabilities message is out of scope for mobility class 1 applications of this profile.

Q.20 Parameters of B_field access request message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	IBCN	7.3.3.2	M		0	
2	Exchanged connection number	8.1.1	M		0 - 15	
3	Logical bearer number	10.2.4	M		0 - 15	
4	Connection type	7.3.3.2	M		3	
5	Service type	7.3.3.2	M		3	
6	Maximum life	7.3.3.2	M		0 - 7	
7	Slot type	7.3.3.2	M		0	

Q.24 Parameters of B_field bearer confirm message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Exchanged connection number	8.1.1	M		0 - 15	
2	Logical bearer number	10.2.4	M		0 - 15	
3	Connection type	7.3.3.3	M		3	
4	Service type	7.3.3.3	M		3	
5	Maximum life	7.3.3.3	M		0 - 7	
6	Slot type	7.3.3.3	M		0	

Q.32 Parameters of B_field bearer release message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Reason for release	7.3.3.10	M		0-3 and 6-13	

Q.33 Parameters of B_field null message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	C_F data indication	7.3.4	C3		0-4	

C.2.6 Functions implemented

Q.34 Function implemented

Supported functions				
Item no.	**Function name**	**Ref.**	**Status**	**Support**
1	B_field data scrambling	6.2.4	M	
2	B_field data unscrambling	6.2.4	M	
3	R-CRC generation	6.2.5.2	M	
4	R-CRC checking	6.2.5.2	M	

(continued)

Q.34 Function implemented (concluded)

Supported functions				
5	X-CRC generation	6.2.5.4	M	
6	X-CRC checking	6.2.5.4	O	
7	Z-CRC generation	ETS 300 175-2 [2] subclause 4.8	I	
8	Z-CRC checking	ETS 300 175-2 [2] subclause 4.8	I	
9	D-MAP D80 field mapping	6.2.1.1	I	
10	D-MAP D32 field mapping	6.2.1.1	M	
11	D-MAP D08 field mapping	6.2.1.1	I	
12	D-MAP D00 field mapping	6.2.1.1	O	
13	A-MAP A_field mapping	6.2.1.2	M	
14	E/U-MUX E80 type multiplex	6.2.2.2	I	
15	E/U-MUX E32 type multiplex	6.2.2.2	M	
16	E/U-MUX E08 type multiplex	6.2.2.2	I	
17	E/U-MUX U80a type multiplex	6.2.2.2	I	
18	E/U-MUX U80b type multiplex	6.2.2.2	I	
19	E/U-MUX U32a type multiplex	6.2.2.2	C2	
20	E/U-MUX U32b type multiplex	6.2.2.2	M	
21	E/U-MUX U08a type multiplex	6.2.2.2	I	
22	E/U-MUX U08b type multiplex	6.2.2.2	I	
23	C-MUX B_field full slot mode 0 multiplex	6.2.2.3.1	M	
24	C-MUX B_field full slot mode 1 multiplex	6.2.2.3.1	C3	
25	C-MUX B_field full slot mode 2 multiplex	6.2.2.3.1	C3	
26	C-MUX B_field full slot mode 3 multiplex	6.2.2.3.1	C3	
27	C-MUX B_field full slot mode 4 multiplex	6.2.2.3.1	M	
28	C-MUX B_field half slot mode 0 multiplex	6.2.2.3.2	I	
29	C-MUX B_field half slot mode 1 multiplex	6.2.2.3.2	I	
30	T-MUX Tail multiplex	6.2.2.1.2	M	
31	Frequency correction function	11.5.2.2	O	

C.2.7 Timer support

Q.35 Timer support

Timer supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	T200	10.2	M		3 seconds	
2	T201	11.5	M		5 seconds	
3	T202	10.6	I		3 seconds	
4	T203	10.6	I		16 frames	
5	T204	9.1	M		6 multiframes	
6	T205	9.1	M		10 seconds	
7	T206	11.2	I		10 frames	
8	T207	11.3	M		5 seconds	
9	T208	11.3	M		20 seconds	
10	T209	11.4	M		30 seconds	
11	T210	11.4	M		2 seconds	

(continued)

Q.35 Timer support (concluded)

Timer supported						
12	T211	10.3	M		3 seconds	
13	T212	10.5	M		20 frames	
14	T213	10.7	M		20 frames	
15	T214	9.2	I		20 frames	
16	T215	9.2	I		6 multiframes	

C.2.8 Procedure support

Q.36 Procedure support

Procedures supported				
Item no.	**Name of procedure**	**Ref.**	**Status**	**Support**
1	Downlink connectionless procedure	9.1.2	C2	
2	Downlink broadcast procedure	9.1.1	M	
3	Uplink connectionless procedure	9.2.3	I	
4	Request for specific Q information procedure	9.3.1	O	
5	Request for a new dummy procedure	9.3.2	O	
6	Basic setup procedure for single bearer basic connection of known service	10.2.4.2	I	
7	Normal setup procedure for single bearer advanced connection of known service	10.2.4.2	M	
8	Fast setup procedure for single bearer advanced connection of known service	10.2.4.2	M	
9	Normal setup procedure multi-bearer symmetric connection	10.2.4.3.1	I	
10	Fast setup procedure for multi-bearer symmetric connection	10.2.4.3.1	I	
11	Normal setup procedure for full asymmetric downlink connection	10.2.4.3.3	I	
12	Fast setup procedure for full asymmetric downlink connection	10.2.4.3.3	I	
13	Normal setup procedure for full asymmetric uplink connection	10.2.4.3.2	I	
14	Fast setup procedure for full asymmetric uplink connection	10.2.4.3.2	I	
15	Setup procedure for connection with unknown service	10.2.4.3	I	
16	Connection modification procedure	10.3	I	
17	Connection release procedure	10.4	M	
18	Basic single bearer setup procedure	10.5.1.1	I	
19	A_field advanced single bearer setup procedure	10.5.1.2	I	
20	B_field single bearer normal setup procedure	10.5.1.3.1	M	
21	B_field single bearer fast setup procedure	10.5.1.3.2	I	
22	Double simplex bearer indirect setup procedure	10.5.1.4	I	
23	Double simplex bearer direct setup procedure	10.5.1.4	I	
24	Duplex bearer handover procedure	10.6.2	I	
25	Double simplex bearer handover procedure	10.6.3	I	
26	Unacknowledged bearer release procedure	10.7.2.1	M	
27	Acknowledged bearer release procedure	10.7.2.2	I	
28	Fast bearer release procedure	10.7.2.3	I	
29	Q1 and Q2 bits setting procedure for C-channel data	10.8.1	C3	
30	Q1 and Q2 bits setting procedure for I_N and I_P_error_detection data	10.8.1.3	I	
31	BCK and Q2 bits setting for I_P_error_correction data in symmetric connection	10.8.2.4.1	M	
32	BCK and ACK bit setting for I_P_error_correction data in asymmetric connection	10.8.2.4.1	I	
33	Bearer replacement procedure	10.8.2.5.1	I	
34	MAC message jump procedure	10.8.2.5.2	M	

(continued)

Q.36 Procedure support (concluded)

Procedures supported				
35	Unilateral jump procedure	10.8.2.5.3	M	
36	Idle_locked state entering procedure	11.3.2	N/A	
37	Idle_locked state maintaining procedure	11.3.3	N/A	
38	Duplex bearer channel selection procedure	11.4.1	M	
39	Double simplex bearer channel selection procedure	11.4.1	I	
40	Simplex bearer channel selection procedure	11.4.1	I	
41	Uplink connectionless channel selection	9.2.2	I	
42	RFPI handshaking procedure	11.5.1	M	
43	PT frequency correction procedure	11.5.2.2	O	
44	MAC layer test message procedure	12.2	M	
45	Receiver scanning procedure	11.9	M	
46	PP paging procedure	9.1.3.2	M	
47	Channel list procedure	10.5.2	I	

Q.37 Parameters of channel selection procedure (duplex/double/simplex bearer)

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Lowest boundary of channel list	11.4.1	M		< = - 93 dBm	
2	Band resolution	11.4.1	M		6 dB	
3	RSSI variation between checking	11.4.1	M		< = 12 dB	

Annex D (normative): Service A: PT Profile ICS Proforma

Notwithstanding the provisions of the copyright clause related to the text of this ETS, ETSI grants that users of this ETS may freely reproduce the PICS proforma in this annex so that it can be used for its intended purposes and may further publish the completed PICS.

D.1 Standardised symbols for the status column

The standardised symbols for the status column are as follows:

- m or M for mandatory;
- o or O for optional (Boolean);
- x or X for prohibited use;
- n/a, N/A or - (dash) for not applicable;
- cn or Cn for conditional on the implementation of feature n;
- i or I for out of scope (the capability is allowed to be implemented but is not called upon by the profile functionality).

Where reception of a message is marked as "O", this shall be understood to mean that correct understanding and processing of that message is optional.

C1: IF Encryption supported THEN M ELSE I.

C2: IF Connectionless downlink supported THEN M ELSE I.

C3: IF Class 2 mobility supported THEN M ELSE I.

Except when stated otherwise, the reference column refers to the relevant subclause in ETS 300 175-3 [3].

D.2 Capabilities

D.2.1 Services

Q.9 Service support

Service supported				
Item no.	**Name of service**	**Ref.**	**Status**	**Support**
1	Type 1f I_N_minimum delay	5.6.2.1	I	
2	Type 2f I_N_normal delay	5.6.2.1	I	
3	Type 3f I_P_error_detection	5.6.2.1	I	
4	Type 4f I_P_error_correction	5.6.2.1	M	
5	Type 5 I_N_normal delay	5.6.2.2	I	
6	Type 6 I_P_error_detection	5.6.2.2	I	
7	Type 7f I_P_error_correction	5.6.2.2	I	
8	Type 1h I_N_minimum delay	5.6.2.1	I	
9	Type 2h I_N_normal delay	5.6.2.1	I	
10	Type 3h I_P_error_detection	5.6.2.1	I	
11	Type 4h I_P_error_correction	5.6.2.1	I	
12	C_S duplex	5.3.1.1	C3	
13	C_F duplex	5.3.1.1	C3	

(continued)

Q.9 Service support (concluded)

Service supported				
Item no.	**Name of service**	**Ref.**	**Status**	**Support**
14	Downlink CL_S only	5.7.2.1	I	
15	Downlink CL_F + CL_S	5.7.2.1	I	
16	Downlink CL_S + SI_N	5.7.2.1	C2	
17	Uplink CL_F only	5.7.2.2	I	
18	Uplink CL_S only	5.7.2.2	I	
19	Uplink without SDU	5.7.2.2	I	
20	G_F simplex	5.3.1.3	M	
21	Fast paging	11.3.3.1	O	
22	Normal paging	11.3.3.1	M	
23	Low duty cycle paging	11.3.3.1	O	
24	System identities	11.3.2	M	
25	System information	11.3.2	M	
26	Encryption	6.2.3	C1	
27	Fast setup	11.3.3.2	I	

D.2.2 Service parameters

Q.10 Service parameters

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	No. of bearers in T2f service	5.6.2.1	I			
2	No. of bearers in T3f service	5.6.2.1	I			
3	No. of bearers in T4f service	5.6.2.1	M		1	
4	Forward bearer no. in T5 service	5.6.2.2	I			
5	Special bearer no. in T5 service	5.6.2.2	I			
6	Forward bearer no. in T6 service	5.6.2.2	I			
7	Special bearer no. in T6 service	5.6.2.2	I			
8	Forward bearer no. in T7 service	5.6.2.2	I			
9	Special bearer no. in T7 service	5.6.2.2	I			

D.2.3 Protocol parameters

Q.11 Protocol parameters

Protocol parameters supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Value**	
					Allowed	**Supported**
1	N200	10.2	M		10	
2	N201	10.6	I			
3	N202	11.4	M		10	
4	N203	9.2	I			

D.2.4 Messages

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
1	TA code: CT data packet number 0	7.1.2	C3		C3	
2	TA code: CT data packet number 1	7.1.2	C3		C3	
3	TA code: identities information on connectionless bearer	7.1.2	N/A		C2	
4	TA code: identities information	7.1.2	M		M	
5	TA code: multiframe sync. and system information	7.1.2	N/A		M	
6	TA code: main escape	7.1.2	X		I	
7	TA code: MAC layer control	7.1.2	M		M	
8	TA code: paging/first PT transmission	7.1.2	M		M	
9	BA code: U-type, I_N, SI_N or I_P packet number 0	7.1.4	M		M	
10	BA code: U-type, I_P error detect or I_P packet number 1	7.1.4	M		M	
11	BA code: E-type, all C_F or CL_F, packet number 0	7.1.4	C3		C3	
12	BA code: E-type, all C_F , packet number 1	7.1.4	C3		C3	
13	BA code: E-type, not all C_F or CL_F; C_F packet number 0	7.1.4	C3		C3	
14	BA code: E-type, not all C_F ; C_F packet number 1	7.1.4	C3		C3	
15	BA code: E-type, all MAC control	7.1.4	M		M	
16	BA code: no B_field	7.1.4	N/A		O	
17	NT information	7.2.2	M		M	
18	Static system information	7.2.3.2	N/A		M	
19	Extended RF carrier information	7.2.3.3	N/A		I	
20	Fixed part capabilities	7.2.3.4	N/A		M	
21	Extended fixed part capabilities	7.2.3.5	N/A		I	
22	SARI message	7.2.3.6	N/A		O	
23	Multi-frame number	7.2.3.7	N/A		C1	
24	Q-channel escape	7.2.3.8	N/A		I	
25	Zero length page indication	7.2.4.2.3	N/A		M	
26	Short page indication	7.2.4.2.3	N/A		M	
27	Full page indication	7.2.4.2.3	N/A		O	
28	Not the last 36 bits of a long page indication	7.2.4.2.3	N/A		I	
29	The first 36 bits of a long page indication	7.2.4.2.3	N/A		I	
30	The last 36 bits of a long page indication	7.2.4.2.3	N/A		I	
31	All of a long page indication	7.2.4.2.3	N/A		I	
32	PT header extend flag	7.2.4.2.2	N/A		M	
33	PT MAC information: fill	7.2.4.3.2	N/A		M	
34	PT MAC information: blind full slot	7.2.4.3.3	N/A		M	
35	PT MAC information: other bearer	7.2.4.3.4	N/A		O	
36	PT MAC information: recommended other bearer	7.2.4.3.4	N/A		O	
37	PT MAC information: good RFP bearer	7.2.4.3.4	N/A		O	
38	PT MAC information: dummy or C/L bearer position	7.2.4.3.4	N/A		M	
39	PT MAC information: RFP identity	7.2.4.3.5	N/A		I	
40	PT MAC information: dummy or C/L bearer marker	7.2.4.3.7	N/A		C2	
41	PT MAC information: escape	7.2.4.3.6	N/A		I	
42	PT MAC information: bearer handover type 0000	7.2.4.3.8	N/A		I	
43	PT MAC information: bearer handover type 0001	7.2.4.3.8	N/A		I	
44	PT MAC information: bearer handover type 0010	7.2.4.3.8	N/A		I	
45	PT MAC information: bearer handover type 0011	7.2.4.3.8	N/A		I	
46	Basic access request	7.2.5.2	I		N/A	

(continued)

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
47	Basic bearer handover request	7.2.5.2	I		N/A	
48	Basic connection handover request	7.2.5.2	I		N/A	
49	Basic unconfirmed access request	7.2.5.2	I		N/A	
50	Basic bearer confirm	7.2.5.2	N/A		I	
51	Basic wait	7.2.5.2.3	I		I	
52	Basic bearer release	7.2.5.2	I		I	
53	Advanced access request	7.2.5.3.2	I		I	
54	Advanced bearer handover request	7.2.5.3.3	I		I	
55	Advanced connection handover request	7.2.5.3.4	I		I	
56	Advanced bearer confirm	7.2.5.3.6	I		I	
57	Advanced unconfirmed access request	7.2.5.3.5	I		I	
58	Advanced wait	7.2.5.3.7	I		I	
59	Advanced attributes-T request	7.2.5.3.8	I		I	
60	Advanced attributes-T confirm	7.2.5.3.8	I		I	
61	Advanced bandwidth-T request	7.2.5.3.9	I		I	
62	Advanced bandwidth-T confirm	7.2.5.3.9	I		I	
63	Advanced channel list	7.2.5.3.10	I		I	
64	Advanced unconfirmed dummy	7.2.5.3.11	I		I	
65	Advanced unconfirmed handover	7.2.5.3.12	I		I	
66	Advanced bearer release	7.2.5.3.13	I		I	
67	Transmit on the specified slot/frequency test message	7.2.5.4.2	N/A		M	
68	Loopback B_field test message	7.2.5.4.3	N/A		M	
69	Defeat antenna diversity test message	7.2.5.4.4	N/A		M	
70	Force bearer handover test message	7.2.5.4.5	N/A		I	
71	Clear test mode test message	7.2.5.4.6	N/A		M	
72	Single bearer antenna switch QC message	7.2.5.5	O		O	
73	All bearer antenna switch QC message	7.2.5.5	O		O	
74	Bearer handover QC message	7.2.5.5	I		I	
75	Connection handover QC message	7.2.5.5	I		I	
76	Single bearer frequency control QC message	7.2.5.5	O		O	
77	All bearer frequency control QC message	7.2.5.5	O		O	
78	C/L single transmission with 1 CL_F segment	7.2.5.6	I		I	
79	C/L single transmission with 2 CL_F segments	7.2.5.6	I		I	
80	C/L single transmission with 3 CL_F segments	7.2.5.6	I		I	
81	C/L single transmission with 4 CL_F segments	7.2.5.6	I		I	
82	1st C/L half slot transmission with 1 CL_F segment	7.2.5.6	I		I	
83	1st C/L full slot transmission with 4 CL_F segment	7.2.5.6	I		I	
84	2nd C/L transmission with 1 CL_F segment	7.2.5.6	I		I	
85	2nd C/L transmission with 2 CL_F segment	7.2.5.6	I		I	
86	2nd C/L transmission with 3 CL_F segment	7.2.5.6	I		I	
87	2nd C/L transmission with 4 CL_F segment	7.2.5.6	I		I	
88	C/L single transmission without CL_F segment	7.2.5.6	I		I	
89	1st C/L transmission without CL_F segment (CL_S service)	7.2.5.6	I		I	
90	Change dummy position	7.2.5.6	O		O	
91	Extended system information	7.2.5.6	O		O	
92	Start encryption: request	7.2.5.7	C1		N/A	

(continued)

Q.12 Messages in A_tail (concluded)

Supported messages						
93	Start encryption: confirm	7.2.5.7	N/A		C1	
94	Start encryption: grant	7.2.5.7	C1		N/A	
95	Stop encryption: request	7.2.5.7	C1		N/A	
96	Stop encryption: confirm	7.2.5.7	N/A		C1	
97	Stop encryption: grant	7.2.5.7	C1		N/A	
98	First transmission for B_field setup	7.2.5.8	M		M	
99	M-channel escape	7.2.5.9	I		I	

Q.13 Messages in B_field

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
1	B_field access request	7.3.3.2	M		I	
2	B_field bearer handover request	7.3.3.2	I		I	
3	B_field connection handover request	7.3.3.2	I		I	
4	B_field unconfirmed access request	7.3.3.2	I		I	
5	B_field bearer confirm	7.3.3.3	I		M	
6	B_field wait	7.3.3.4	M		M	
7	B_field attributes-B request	7.3.3.5	I		I	
8	B_field attributes-B confirm	7.3.3.5	I		I	
9	B_field bandwidth-B request	7.3.3.6	I		I	
10	B_field bandwidth-B confirm	7.3.3.6	I		I	
11	B_field channel list	7.3.3.7	I		I	
12	B_field unconfirmed dummy	7.3.3.8	I		I	
13	B_field unconfirmed handover	7.3.3.9	I		I	
14	B_field bearer release	7.3.3.10	M		M	
15	B_field null message	7.3.4	M		M	
16	B_field single bearer antenna switch QC message	7.3.5.2	O		O	
17	B_field all bearer antenna switch QC message	7.3.5.2	O		O	
18	B_field bearer handover QC message	7.3.5.2	I		I	
19	B_field connection handover QC message	7.3.5.2	I		I	
20	B_field single bearer frequency control QC message	7.3.5.2	O		O	
21	B_field all bearer frequency control QC message	7.3.5.2	O		O	
22	B-field reset request	7.3.5.3	M		M	
23	B_field reset confirm	7.3.5.3	M		M	
24	MAC-MOD2-ACK message	7.3.5.4	M		M	
25	B-field request for QT field	7.3.6.2	O		N/A	
26	B-field QT field response	7.3.6.3	N/A		O	
27	B_field TARI message	7.3.6.4	N/A		O	
28	B_field G_F data	7.3.7	M		M	
29	B_field M-channel escape	7.3.8	O		O	

D.2.5 Message parameters

Q.14 Parameters of static system information message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Normal-reverse	7.2.3.2.2	M		0 - 1	
2	Slot number	7.2.3.2.3	M		0 - 11	
3	Start position	7.2.3.2.4	M		0	
4	Escape	7.2.3.2.5	M		0 - 1	
5	Number of transceivers	7.2.3.2.6	M		0 - 3	
6	Extended RF carrier flag	7.2.3.2.7	M		0	
7	RF carriers available	7.2.3.2.8	M		1 - 1 023	
8	Spare	7.2.3.2.11	M		0	
9	Carrier number	7.2.3.2.10	M		0 - 9	
10	Spare	7.2.3.2.11	M		0	
11	Primary receiver scan number	7.2.3.2.12	M		0 - 9	

Q.15 Parameters of FP capabilities message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Extended FP information	7.2.3.4.2	M		0, 1	
2	Full slot	7.2.3.4.2	M		1	
3	Frequency control	7.2.3.4.2	M		0, 1	
4	Page repetition	7.2.3.4.2	M		0, 1	
5	C/O setup on dummy allowed	7.2.3.4.2	M		C2 = 0	
6	C/L uplink	7.2.3.4.2	M		0, 1	
7	C/L downlink	7.2.3.4.2	M		C2 = 1	
8	Basic A_field setup	7.2.3.4.2	M		0, 1	
9	Advanced A_field setup	7.2.3.4.2	M		0, 1	
10	B_field setup	7.2.3.4.2	M		1	
11	C_F messages	7.2.3.4.2	C3		0, 1	
12	I_N_minimum_delay	7.2.3.4.2	M		0, 1	
13	I_N_normal_delay	7.2.3.4.2	M		0, 1	
14	I_P_error_detection	7.2.3.4.2	M		0, 1	
15	I_P_error_correction	7.2.3.4.2	M		1	
16	Multibearer connection	7.2.3.4.2	M		0, 1	

The higher layer information field of the FP capabilities message is out of scope for mobility class 1 applications of this profile.

Q.20 Parameters of B_field access request message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	IBCN	7.3.3.2	M		0, 3	
2	Exchanged connection number	8.1.1	M		0 - 15	

(continued)

Q.20 Parameters of B_field access request message (concluded)

Supported parameters						
3	Logical bearer number	10.2.4	M		0 - 15	
4	Connection type	7.3.3.2	M		3	
5	Service type	7.3.3.2	M		3	
6	Maximum life	7.3.3.2	M		0 - 7	
7	Slot type	7.3.3.2	M		0	

Q.24 Parameters of B_field bearer confirm message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Exchanged connection number	8.1.1	M		0 - 15	
2	Logical bearer number	10.2.4	M		0 - 15	
3	Connection type	7.3.3.3	M		3	
4	Service type	7.3.3.3	M		3	
5	Maximum life	7.3.3.3	M		0 - 7	
6	Slot type	7.3.3.3	M		0	

Q.32 Parameters of B_field bearer release message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Reason for release	7.3.3.10	M		0-3 and 6-13	

Q.33 Parameters of B_field null message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	C_F data indication	7.3.4	C3		0-4	

D.2.6 Functions implemented

Q.34 Function implemented

Item no.	**Function name**	**Ref.**	**Status**	**Support**
1	B_field data scrambling	6.2.4	M	
2	B_field data unscrambling	6.2.4	M	
3	R-CRC generation	6.2.5.2	M	
4	R-CRC checking	6.2.5.2	M	
5	X-CRC generation	6.2.5.4	M	
6	X-CRC checking	6.2.5.4	O	
7	Z-CRC generation	ETS 300 175-2 [2] subclause 4.8	I	
8	Z-CRC checking	ETS 300 175-2 [2] subclause 4.8	I	

(continued)

Q.34 Function implemented (concluded)

9	D-MAP D80 field mapping	6.2.1.1	I	
10	D-MAP D32 field mapping	6.2.1.1	M	
11	D-MAP D08 field mapping	6.2.1.1	I	
12	D-MAP D00 field mapping	6.2.1.1	O	
13	A-MAP A_field mapping	6.2.1.2	M	
14	E/U-MUX E80 type multiplex	6.2.2.2	I	
15	E/U-MUX E32 type multiplex	6.2.2.2	M	
16	E/U-MUX E08 type multiplex	6.2.2.2	I	
17	E/U-MUX U80a type multiplex	6.2.2.2	I	
18	E/U-MUX U80b type multiplex	6.2.2.2	I	
19	E/U-MUX U32a type multiplex	6.2.2.2	C2	
20	E/U-MUX U32b type multiplex	6.2.2.2	M	
21	E/U-MUX U08a type multiplex	6.2.2.2	I	
22	E/U-MUX U08b type multiplex	6.2.2.2	I	
23	C-MUX B_field full slot mode 0 multiplex	6.2.2.3.1	M	
24	C-MUX B_field full slot mode 1 multiplex	6.2.2.3.1	C3	
25	C-MUX B_field full slot mode 2 multiplex	6.2.2.3.1	C3	
26	C-MUX B_field full slot mode 3 multiplex	6.2.2.3.1	C3	
27	C-MUX B_field full slot mode 4 multiplex	6.2.2.3.1	M	
28	C-MUX B_field half slot mode 0 multiplex	6.2.2.3.2	I	
29	C-MUX B_field half slot mode 1 multiplex	6.2.2.3.2	I	
30	T-MUX Tail multiplex	6.2.2.1.2	M	
31	Frequency correction function	11.5.2.2	O	

D.2.7 Timer support

Q.35 Timer support

Timer supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	T200	10.2	M		3 seconds	
2	T201	11.5	M		5 seconds	
3	T202	10.6	I		3 seconds	
4	T203	10.6	I		16 frames	
5	T204	9.1	M		6 multiframes	
6	T205	9.1	M		10 seconds	
7	T206	11.2	I		10 frames	
8	T207	11.3	M		5 seconds	
9	T208	11.3	M		20 seconds	
10	T209	11.4	M		30 seconds	
11	T210	11.4	M		2 seconds	
12	T211	10.3	M		3 seconds	
13	T212	10.5	M		20 frames	
14	T213	10.7	M		20 frames	
15	T214	9.2	I		20 frames	
16	T215	9.2	I		6 multiframes	

D.2.8 Procedure support

Q.36 Procedure support

Procedures supported				
Item no.	**Name of procedure**	**Ref.**	**Status**	**Support**
1	Downlink connectionless procedure	9.1.2	C2	
2	Downlink broadcast procedure	9.1.1	M	
3	Uplink connectionless procedure	9.2.3	I	
4	Request for specific Q information procedure	9.3.1	O	
5	Request for a new dummy procedure	9.3.2	O	
6	Basic setup procedure for single bearer basic connection of known service	10.2.4.2	I	
7	Normal setup procedure for single bearer advanced connection of known service	10.2.4.2	M	
8	Fast setup procedure for single bearer advanced connection of known service	10.2.4.2	I	
9	Normal setup procedure for multi-bearer symmetric connection	10.2.4.3.1	I	
10	Fast setup procedure for multi-bearer symmetric connection	10.2.4.3.1	I	
11	Normal setup procedure for full asymmetric downlink connection	10.2.4.3.3	I	
12	Fast setup procedure for full asymmetric downlink connection	10.2.4.3.3	I	
13	Normal setup procedure for full asymmetric uplink connection	10.2.4.3.2	I	
14	Fast setup procedure for full asymmetric uplink connection	10.2.4.3.2	I	
15	Setup procedure for connection with unknown service	10.2.4.3	I	
16	Connection modification procedure	10.3	I	
17	Connection release procedure	10.4	M	
18	Basic single bearer setup procedure	10.5.1.1	I	
19	A_field advanced single bearer setup procedure	10.5.1.2	I	
20	B_field single bearer normal setup procedure	10.5.1.3.1	M	
21	B_field single bearer fast setup procedure	10.5.1.3.2	I	
22	Double simplex bearer indirect setup procedure	10.5.1.4	I	
23	Double simplex bearer direct setup procedure	10.5.1.4	I	
24	Duplex bearer handover procedure	10.6.2	I	
25	Double simplex bearer handover procedure	10.6.3	I	
26	Unacknowledged bearer release procedure	10.7.2.1	M	
27	Acknowledged bearer release procedure	10.7.2.2	I	
28	Fast bearer release procedure	10.7.2.3	I	
29	Q1 and Q2 bits setting procedure for C-channel data	10.8.1	C3	
30	Q1 and Q2 bits setting procedure for I_N and I_P_error_detection data	10.8.1.3	I	
31	BCK and Q2 bits setting for I_P_error_correction data in symmetric connection	10.8.2.4.1	M	
32	BCK and ACK bit setting for I_P_error_correction data in asymmetric connection	10.8.2.4.1	I	
33	Bearer replacement procedure	10.8.2.5.1	I	
34	MAC message jump procedure	10.8.2.5.2	M	
35	Unilateral jump procedure	10.8.2.5.3	M	
36	Idle_locked state entering procedure	11.3.2	N/A	
37	Idle_locked state maintaining procedure	11.3.3	N/A	
38	Duplex bearer channel selection procedure	11.4.1	M	
39	Double simplex bearer channel selection procedure	11.4.1	I	
40	Simplex bearer channel selection procedure	11.4.1	I	
41	Uplink connectionless channel selection	9.2.2	I	
42	RFPI handshaking procedure	11.5.1	M	
43	PT frequency correction procedure	11.5.2.2	O	

(continued)

Q.36 Procedure support (concluded)

44	MAC layer test message procedure	12.2	M	
45	Receiver scanning procedure	11.9	M	
46	PP paging procedure	9.1.3.2	M	
47	Channel list procedure	10.5.2	I	

Q.37 Parameters of channel selection procedure (duplex/double/simplex bearer)

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Lowest boundary of channel list	11.4.1	M		< = - 93 dBm	
2	Band resolution	11.4.1	M		6 dB	
3	RSSI variation between checking	11.4.1	M		< = 12 dB	

Annex E (normative): Service B: FT Profile ICS Proforma

Notwithstanding the provisions of the copyright clause related to the text of this ETS, ETSI grants that users of this ETS may freely reproduce the PICS proforma in this annex so that it can be used for its intended purposes and may further publish the completed PICS.

E.1 Standardised symbols for the status column

The standardised symbols for the status column are as follows:

- m or M for mandatory;
- o or O for optional (Boolean);
- x or X for prohibited use;
- n/a, N/A or - (dash) for not applicable;
- cn or Cn for conditional on the implementation of feature n;
- i or I for out of scope (the capability is allowed to be implemented but is not called upon by the profile functionality).

Where reception of a message is marked as "O", this shall be understood to mean that correct understanding and processing of that message is optional.

C1: IF Encryption supported THEN M ELSE I.

C2: IF Connectionless downlink supported THEN M ELSE I.

C3: IF Class 2 mobility supported THEN M ELSE I.

Except when stated otherwise, the reference column refers to the relevant subclause in ETS 300 175-3 [3].

E.2 Capabilities

E.2.1 Services

Q.9 Service support

Service supported				
Item no.	**Name of service**	**Ref.**	**Status**	**Support**
1	Type 1f I_N_minimum delay	5.6.2.1	I	
2	Type 2f I_N_normal delay	5.6.2.1	I	
3	Type 3f I_P_error_detection	5.6.2.1	I	
4	Type 4f I_P_error_correction	5.6.2.1	M	
5	Type 5 I_N_normal delay	5.6.2.2	I	
6	Type 6 I_P_error_detection	5.6.2.2	I	
7	Type 7f I_P_error_correction	5.6.2.2	M	
8	Type 1h I_N_minimum delay	5.6.2.1	I	
9	Type 2h I_N_normal delay	5.6.2.1	I	
10	Type 3h I_P_error_detection	5.6.2.1	I	
(continued)				

Q.9 Service support (concluded)

Service supported				
11	Type 4h I_P_error_correction	5.6.2.1	I	
12	C_S duplex	5.3.1.1	C3	
13	C_F duplex	5.3.1.1	C3	
14	Downlink CL_S only	5.7.2.1	I	
15	Downlink CL_F + CL_S	5.7.2.1	I	
16	Downlink CL_S+ SIN	5.7.2.1	C2	
17	Uplink CL_F only	5.7.2.2	I	
18	Uplink CL_Sonly	5.7.2.2	I	
19	Uplink without SDU	5.7.2.2	I	
20	G_F simplex	5.3.1.3	M	
21	Fast paging	11.3.3.1	O	
22	Normal paging	11.3.3.1	M	
23	Low duty cycle paging	11.3.3.1	O	
24	System identities	11.3.2	M	
25	System information	11.3.2	M	
26	Encryption	6.2.3	C1	
27	Fast setup	11.3.3.2	M	

E.2.2 Service parameters

Q.10 Service parameters

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	No. of bearers in T2f service	5.6.2.1	I			
2	No. of bearers in T3f service	5.6.2.1	I			
3	No. of bearers in T4f service	5.6.2.1	M		1 - 12	
4	Forward bearer no. in T5 service	5.6.2.2	I			
5	Special bearer no. in T5 service	5.6.2.2	I			
6	Forward bearer no. in T6 service	5.6.2.2	I			
7	Special bearer no. in T6 service	5.6.2.2	I			
8	Forward bearer no. in T7 service	5.6.2.2	M		1 - 23	
9	Special bearer no. in T7 service	5.6.2.2	M		1	

E.2.3 Protocol parameters

Q.11 Protocol parameters

Protocol parameters supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Value**	
					Allowed	**Supported**
1	N200	10.2	M		10	
2	N201	10.6	I			
3	N202	11.4	M		10	
4	N203	9.2	I			

E.2.4 Messages

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	Support	Status	Support
1	TA code: CT data packet number 0	7.1.2	C3		C3	
2	TA code: CT data packet number 1	7.1.2	C3		C3	
3	TA code: identities information on connectionless bearer	7.1.2	C2		N/A	
4	TA code: identities information	7.1.2	M		M	
5	TA code: multiframe sync. and system information	7.1.2	M		N/A	
6	TA code: main escape	7.1.2	X		I	
7	TA code: MAC layer control	7.1.2	M		M	
8	TA code: paging/first PT transmission	7.1.2	M		M	
9	BA code: U-type, I_N, SI_N or I_P packet number 0	7.1.4	M		M	
10	BA code: U-type, I_P error detect or I_P packet number 1	7.1.4	M		M	
11	BA code: E-type, all C_F or CL_F, packet number 0	7.1.4	C3		C3	
12	BA code: E-type, all C_F , packet number 1	7.1.4	C3		C3	
13	BA code: E-type, not all C_F or CL_F; C_F packet number 0	7.1.4	C3		C3	
14	BA code: E-type, not all C_F; C_F packet number 1	7.1.4	C3		C3	
15	BA code: E-type, all MAC control	7.1.4	M		M	
16	BA code: no B_field	7.1.4	O		N/A	
17	NT information	7.2.2	M		M	
18	Static system information	7.2.3.2	M		N/A	
19	Extended RF carrier information	7.2.3.3	I		N/A	
20	Fixed part capabilities	7.2.3.4	M		N/A	
21	Extended fixed part capabilities	7.2.3.5	I		N/A	
22	SARI message	7.2.3.6	O		N/A	
23	Multi-frame number	7.2.3.7	C1		N/A	
24	Q-channel escape	7.2.3.8	I		N/A	
25	Zero length page indication	7.2.4.2.3	M		N/A	
26	Short page indication	7.2.4.2.3	M		N/A	
27	Full page indication	7.2.4.2.3	I		N/A	
28	Not the last 36 bits of a long page indication	7.2.4.2.3	I		N/A	
29	The first 36 bits of a long page indication	7.2.4.2.3	I		N/A	
30	The last 36 bits of a long page indication	7.2.4.2.3	I		N/A	
31	All of a long page indication	7.2.4.2.3	I		N/A	
32	PT header extend flag	7.2.4.2.2	M		N/A	
33	PT MAC information: fill	7.2.4.3.2	O		N/A	
34	PT MAC information: blind full slot	7.2.4.3.3	M		N/A	
35	PT MAC information: other bearer	7.2.4.3.4	O		N/A	
36	PT MAC information: recommended other bearer	7.2.4.3.4	O		N/A	
37	PT MAC information: good RFP bearer	7.2.4.3.4	O		N/A	
38	PT MAC information: dummy or C/L bearer position	7.2.4.3.4	M		N/A	
39	PT MAC information: RFP identity	7.2.4.3.5	I		N/A	
40	PT MAC information: dummy or C/L bearer marker	7.2.4.3.7	C2		N/A	
41	PT MAC information: escape	7.2.4.3.6	I		N/A	
42	PT MAC information: bearer handover type 0000	7.2.4.3.8	I		N/A	

(continued)

Q.12 Messages in A_tail

Supported messages						
44	PT MAC information: bearer handover type 0010	7.2.4.3.8	I		N/A	
45	PT MAC information: bearer handover type 0011	7.2.4.3.8	I		N/A	
46	Basic access request	7.2.5.2	N/A		I	
47	Basic bearer handover request	7.2.5.2	N/A		I	
48	Basic connection handover request	7.2.5.2	N/A		I	
49	Basic unconfirmed access request	7.2.5.2	N/A		I	
50	Basic bearer confirm	7.2.5.2	I		N/A	
51	Basic wait	7.2.5.2.3	I		I	
52	Basic bearer release	7.2.5.2	I		I	
53	Advanced access request	7.2.5.3.2	I		I	
54	Advanced bearer handover request	7.2.5.3.3	I		I	
55	Advanced connection handover request	7.2.5.3.4	I		I	
56	Advanced bearer confirm	7.2.5.3.6	I		I	
57	Advanced unconfirmed access request	7.2.5.3.5	I		I	
58	Advanced wait	7.2.5.3.7	I		I	
59	Advanced attributes-T request	7.2.5.3.8	I		I	
60	Advanced attributes-T confirm	7.2.5.3.8	I		I	
61	Advanced bandwidth-T request	7.2.5.3.9	I		I	
62	Advanced bandwidth-T confirm	7.2.5.3.9	I		I	
63	Advanced channel list	7.2.5.3.10	I		I	
64	Advanced unconfirmed dummy	7.2.5.3.11	I		I	
65	Advanced unconfirmed handover	7.2.5.3.12	I		I	
66	Advanced bearer release	7.2.5.3.13	I		I	
67	Transmit on the specified slot/frequency test message	7.2.5.4.2	N/A		N/A	
68	Loopback B_field test message	7.2.5.4.3	N/A		M	
69	Defeat antenna diversity test message	7.2.5.4.4	N/A		M	
70	Force bearer handover test message	7.2.5.4.5	N/A		I	
71	Clear test mode test message	7.2.5.4.6	N/A		M	
72	Single bearer antenna switch QC message	7.2.5.5	O		O	
73	All bearer antenna switch QC message	7.2.5.5	O		O	
74	Bearer handover QC message	7.2.5.5	I		I	
75	Connection handover QC message	7.2.5.5	I		I	
76	Single bearer frequency control QC message	7.2.5.5	O		O	
77	All bearer frequency control QC message	7.2.5.5	O		O	
78	C/L single transmission with 1 CL_F segment	7.2.5.6	I		I	
79	C/L single transmission with 2 CL_F segments	7.2.5.6	I		I	
80	C/L single transmission with 3 CL_F segments	7.2.5.6	I		I	
81	C/L single transmission with 4 CL_F segments	7.2.5.6	I		I	
82	1st C/L half slot transmission with 1 CL_F segment	7.2.5.6	I		I	
83	1st C/L full slot transmission with 4 CL_F segment	7.2.5.6	I		I	
84	2nd C/L transmission with 1 CL_F segment	7.2.5.6	I		I	
85	2nd C/L transmission with 2 CL_F segment	7.2.5.6	I		I	
86	2nd C/L transmission with 3 CL_F segment	7.2.5.6	I		I	
87	2nd C/L transmission with 4 CL_F segment	7.2.5.6	I		I	
88	C/L single transmission without CL_F segment	7.2.5.6	I		I	
89	1st C/L transmission without CL_F segment (CL_S service)	7.2.5.6	I		I	

(continued)

Q.12 Messages in A_tail (concluded)

Supported messages						
90	Change dummy position	7.2.5.6	O		O	
91	Extended system information	7.2.5.6	O		O	
92	Start encryption: request	7.2.5.7	N/A		C1	
93	Start encryption: confirm	7.2.5.7	C1		N/A	
94	Start encryption: grant	7.2.5.7	N/A		C1	
95	Stop encryption: request	7.2.5.7	N/A		C1	
96	Stop encryption: confirm	7.2.5.7	C1		N/A	
97	Stop encryption: grant	7.2.5.7	N/A		C1	
98	First transmission for B_field setup	7.2.5.8	M		M	
99	M-channel escape	7.2.5.9	I		I	

Q.13 Messages in B_field

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	**Support**	**Status**	**Support**
1	B_field access request	7.3.3.2	M		M	
2	B_field bearer handover request	7.3.3.2	I		I	
3	B_field connection handover request	7.3.3.2	I		I	
4	B_field unconfirmed access request	7.3.3.2	M		M	
5	B_field bearer confirm	7.3.3.3	M		M	
6	B_field wait	7.3.3.4	M		M	
7	B_field attributes-B request	7.3.3.5	I		I	
8	B_field attributes-B confirm	7.3.3.5	I		I	
9	B_field bandwidth-B request	7.3.3.6	M		M	
10	B_field bandwidth-B confirm	7.3.3.6	M		M	
11	B_field channel list	7.3.3.7	M		M	
12	B_field unconfirmed dummy	7.3.3.8	I		I	
13	B_field unconfirmed handover	7.3.3.9	I		I	
14	B_field bearer release	7.3.3.10	M		M	
15	B_field null message	7.3.4	M		M	
16	B_field single bearer antenna switch QC message	7.3.5.2	O		O	
17	B_field all bearer antenna switch QC message	7.3.5.2	O		O	
18	B_field bearer handover QC message	7.3.5.2	I		I	
19	B_field connection handover QC message	7.3.5.2	I		I	
20	B_field single bearer frequency control QC message	7.3.5.2	O		O	
21	B_field all bearer frequency control QC message	7.3.5.2	O		O	
22	B-field reset request	7.3.5.3	M		M	
23	B_field reset confirm	7.3.5.3	M		M	
24	MAC-MOD2-ACK message	7.3.5.4	M		M	
25	B-field request for QT field	7.3.6.2	N/A		O	
26	B-field QT field response	7.3.6.3	O		N/A	
27	B_field TARI message	7.3.6.4	O		N/A	
28	B_field G_F data	7.3.7	M		M	
29	B_field M-channel escape	7.3.8	O		O	

E.2.5 Message parameters

Q.14 Parameters of static system information message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Normal-reverse	7.2.3.2.2	M		0 - 1	
2	Slot number	7.2.3.2.3	M		0 - 11	
3	Start position	7.2.3.2.4	M		0	
4	Escape	7.2.3.2.5	M		0 - 1	
5	Number of transceivers	7.2.3.2.6	M		0 - 3	
6	Extended RF carrier flag	7.2.3.2.7	M		0	
7	RF carriers available	7.2.3.2.8	M		1 - 1 023	
8	Spare	7.2.3.2.11	M		0	
9	Carrier number	7.2.3.2.10	M		0 - 9	
10	Spare	7.2.3.2.11	M		0	
11	Primary receiver scan number	7.2.3.2.12	M		0 - 9	

Q.15 Parameters of FP capabilities message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Extended FP information	7.2.3.4.2	M		0, 1	
2	Full slot	7.2.3.4.2	M		1	
3	Frequency control	7.2.3.4.2	M		0, 1	
4	Page repetition	7.2.3.4.2	M		0, 1	
5	C/O setup on dummy allowed	7.2.3.4.2	M		C2 = 0	
6	C/L uplink	7.2.3.4.2	M		0, 1	
7	C/L downlink	7.2.3.4.2	M		C2 = 1	
8	Basic A_field setup	7.2.3.4.2	M		0, 1	
9	Advanced A_field setup	7.2.3.4.2	M		0, 1	
10	B_field setup	7.2.3.4.2	M		1	
11	C_F messages	7.2.3.4.2	C3		0, 1	
12	I_N_minimum_delay	7.2.3.4.2	M		0, 1	
13	I_N_normal_delay	7.2.3.4.2	M		0, 1	
14	I_P_error_detection	7.2.3.4.2	M		0, 1	
15	I_P_error_correction	7.2.3.4.2	M		1	
16	Multibearer connection	7.2.3.4.2	M		1	

The higher layer information field of the FP capabilities message is out of scope for mobility class 1 applications of this profile.

Q.20 Parameters of B_field access request message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	IBCN	7.3.3.2	M		0, 3	
2	Exchanged connection number	8.1.1	M		0 - 15	
3	Logical bearer number	10.2.4	M		0 - 15	

(continued)

Q.20 Parameters of B_field access request message (concluded)

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
4	Connection type	7.3.3.2	M		0 - 3	
5	Service type	7.3.3.2	M		3	
6	Maximum life	7.3.3.2	M		0 - 7	
7	Slot type	7.3.3.2	M		0	

Q.24 Parameters of B_field bearer confirm message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Exchanged connection number	8.1.1	M		0 - 15	
2	Logical bearer number	10.2.4	M		0 - 15	
3	Connection type	7.3.3.3	M		0 - 3	
4	Service type	7.3.3.3	M		3	
5	Maximum life	7.3.3.3	M		0 - 7	
6	Slot type	7.3.3.3	M		0	

Q.27 Parameters of B_field Bandwidth-B Request Message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Minimum number of uplink bearers	7.3.3.6	M		1 - 12	
2	Target number of uplink bearers	7.3.3.6	M		1 - 12	
3	Minimum number of downlink bearers	7.3.3.6	M		1 - 12	
4	Target number of downlink bearers	7.3.3.6	M		1 - 12	

Q.28 Parameters of B_field bandwidth-B confirm message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Minimum number of uplink bearers	7.3.3.6	M		1 - 12	
2	Target number of uplink bearers	7.3.3.6	M		1 - 12	
3	Minimum number of downlink bearers	7.3.3.6	M		1 - 12	
4	Target number of downlink bearers	7.3.3.6	M		1 - 12	

Q.29 Parameters of B_field channel list message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	First message type	7.3.3.7	M		0 - 7	
2	Second message type	7.3.3.7	M		0 - 7	
3	Third message type	7.3.3.7	M		0 - 7	

Q.32 Parameters of B_field bearer release message

Supported parameters						
Item no.	Parameter	Ref.	Status	Support	Values	
					Allowed	Supported
1	Reason for release	7.3.3.10	M		0-3 and 6-13	

Q.33 Parameters of B_field null message

Supported parameters						
Item no.	Parameter	Ref.	Status	Support	Values	
					Allowed	Supported
1	C_F data indication	7.3.4	C3		0-4	

E.2.6 Functions implemented

Q.34 Function implemented

Supported functions				
Item no.	Function name	Ref.	Status	Support
1	B_field data scrambling	6.2.4	M	
2	B_field data unscrambling	6.2.4	M	
3	R-CRC generation	6.2.5.2	M	
4	R-CRC checking	6.2.5.2	M	
5	X-CRC generation	6.2.5.4	M	
6	X-CRC checking	6.2.5.4	O	
7	Z-CRC generation	ETS 300 175-2 [2] subclause 4.8	I	
8	Z-CRC checking	ETS 300 175-2 [2] subclause 4.8	I	
9	D-MAP D80 field mapping	6.2.1.1	I	
10	D-MAP D32 field mapping	6.2.1.1	M	
11	D-MAP D08 field mapping	6.2.1.1	I	
12	D-MAP D00 field mapping	6.2.1.1	O	
13	A-MAP A_field mapping	6.2.1.2	M	
14	E/U-MUX E80 type multiplex	6.2.2.2	I	
15	E/U-MUX E32 type multiplex	6.2.2.2	M	
16	E/U-MUX E08 type multiplex	6.2.2.2	I	
17	E/U-MUX U80a type multiplex	6.2.2.2	I	
18	E/U-MUX U80b type multiplex	6.2.2.2	I	
19	E/U-MUX U32a type multiplex	6.2.2.2	C2	
20	E/U-MUX U32b type multiplex	6.2.2.2	M	
21	E/U-MUX U08a type multiplex	6.2.2.2	I	
22	E/U-MUX U08b type multiplex	6.2.2.2	I	
23	C-MUX B_field full slot mode 0 multiplex	6.2.2.3.1	M	
24	C-MUX B_field full slot mode 1 multiplex	6.2.2.3.1	C3	
25	C-MUX B_field full slot mode 2 multiplex	6.2.2.3.1	C3	
26	C-MUX B_field full slot mode 3 multiplex	6.2.2.3.1	C3	
27	C-MUX B_field full slot mode 4 multiplex	6.2.2.3.1	M	
28	C-MUX B_field half slot mode 0 multiplex	6.2.2.3.2	I	
29	C-MUX B_field half slot mode 1 multiplex	6.2.2.3.2	I	
30	T-MUX Tail multiplex	6.2.2.1.2	M	
31	Frequency correction function	11.5.2.2	O	

E.2.7 Timer support

Q.35 Timer support

Timer supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	T200	10.2	M		3 seconds	
2	T201	11.5	M		5 seconds	
3	T202	10.6	I		3 seconds	
4	T203	10.6	I		16 frames	
5	T204	9.1	M		6 multiframes	
6	T205	9.1	M		10 seconds	
7	T206	11.2	I		10 frames	
8	T207	11.3	M		5 seconds	
9	T208	11.3	M		20 seconds	
10	T209	11.4	M		30 seconds	
11	T210	11.4	M		2 seconds	
12	T211	10.3	M		3 seconds	
13	T212	10.5	M		20 frames	
14	T213	10.7	M		20 frames	
15	T214	9.2	I		20 frames	
16	T215	9.2	I		6 multiframes	

E.2.8 Procedure support

Q.36 Procedure support

Procedures supported				
Item no.	**Name of procedure**	**Ref.**	**Status**	**Support**
1	Downlink connectionless procedure	9.1.2	C2	
2	Downlink broadcast procedure	9.1.1	M	
3	Uplink connectionless procedure	9.2.3	I	
4	Request for specific Q information procedure	9.3.1	O	
5	Request for a new dummy procedure	9.3.2	O	
6	Basic setup procedure for single bearer basic connection of known service	10.2.4.2	I	
7	Normal setup procedure for single bearer advanced connection of known service	10.2.4.2	M	
8	Fast setup procedure for single bearer advanced connection of known service	10.2.4.2	M	
9	Normal setup procedure for multi-bearer symmetric connection	10.2.4.3.1	M	
10	Fast setup procedure for multi-bearer symmetric connection	10.2.4.3.1	M	
11	Normal setup procedure for full asymmetric downlink connection	10.2.4.3.3	M	
12	Fast setup procedure for full asymmetric downlink connection	10.2.4.3.3	M	
13	Normal setup procedure for full asymmetric uplink connection	10.2.4.3.2	M	
14	Fast setup procedure for full asymmetric uplink connection	10.2.4.3.2	M	
15	Setup procedure for connection with unknown service	10.2.4.3	I	
16	Connection modification procedure	10.3	M	
17	Connection release procedure	10.4	M	
18	Basic single bearer setup procedure	10.5.1.1	I	
19	A_field advanced single bearer setup procedure	10.5.1.2	I	
20	B_field single bearer normal setup procedure	10.5.1.3.1	M	
21	B_field single bearer fast setup procedure	10.5.1.3.2	M	
22	Double simplex bearer indirect setup procedure	10.5.1.4	M	
23	Double simplex bearer direct setup procedure	10.5.1.4	M	

(continued)

Q.36 Procedure support (concluded)

24	Duplex bearer handover procedure	10.6.2	I	
25	Double simplex bearer handover procedure	10.6.3	I	
26	Unacknowledged bearer release procedure	10.7.2.1	M	
27	Acknowledged bearer release procedure	10.7.2.2	I	
28	Fast bearer release procedure	10.7.2.3	I	
29	Q1 and Q2 bits setting procedure for C-channel data	10.8.1	C3	
30	Q1 and Q2 bits setting procedure for I_N and I_P_error_detection data	10.8.1.3	I	
31	BCK and Q2 bits setting for I_P_error_correction data in symmetric connection	10.8.2.4.1	M	
32	BCK and ACK bit setting for I_P_error_correction data in asymmetric connection	10.8.2.4.1	M	
33	Bearer replacement procedure	10.8.2.5.1	M	
34	MAC message jump procedure	10.8.2.5.2	M	
35	Unilateral jump procedure	10.8.2.5.3	M	
36	Idle_locked state entering procedure	11.3.2	N/A	
37	Idle_locked state maintaining procedure	11.3.3	N/A	
38	Duplex bearer channel selection procedure	11.4.1	M	
39	Double simplex bearer channel selection procedure	11.4.1	M	
40	Simplex bearer channel selection procedure	11.4.1	M	
41	Uplink connectionless channel selection	9.2.2	I	
42	RFPI handshaking procedure	11.5.1	M	
43	PT frequency correction procedure	11.5.2.2	O	
44	MAC layer test message procedure	12.2	M	
45	Receiver scanning procedure	11.9	M	
46	PP paging procedure	9.1.3.2	M	
47	Channel list procedure	10.5.2	M	

Q.37 Parameters of channel selection procedure (duplex/double/simplex bearer)

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Lowest boundary of channel list	11.4.1	M		< = - 93 dBm	
2	Band resolution	11.4.1	M		6 dB	
3	RSSI variation between checking	11.4.1	M		< = 12 dB	

Annex F (normative): Service B: PT Profile ICS Proforma

Notwithstanding the provisions of the copyright clause related to the text of this ETS, ETSI grants that users of this ETS may freely reproduce the PICS proforma in this annex so that it can be used for its intended purposes and may further publish the completed PICS.

F.1 Standardised symbols for the status column

The standardised symbols for the status column are as follows:

- m or M for mandatory;
- o or O for optional (Boolean);
- x or X for prohibited use;
- n/a, N/A or - (dash) for not applicable;
- cn or Cn for conditional on the implementation of feature n;
- i or I for out of scope (the capability is allowed to be implemented but is not called upon by the profile functionality).

Where reception of a message is marked as "O", this shall be understood to mean that correct understanding and processing of that message is optional.

C1: IF Encryption supported THEN M ELSE I.

C2: IF Connectionless downlink supported THEN M ELSE I.

C3: IF Class 2 mobility supported THEN M ELSE I.

Except when stated otherwise, the reference column refers to the relevant subclause in ETS 300 175-3 [3].

F.2 Capabilities

F.2.1 Services

Q.9 Service support

Service supported				
Item no.	**Name of service**	**Ref.**	**Status**	**Support**
1	Type 1f I_N_minimum delay	5.6.2.1	I	
2	Type 2f I_N_normal delay	5.6.2.1	I	
3	Type 3f I_P_error_detection	5.6.2.1	I	
4	Type 4f I_P_error_correction	5.6.2.1	M	
5	Type 5 I_N_normal delay	5.6.2.2	I	
6	Type 6 I_P_error_detection	5.6.2.2	I	
7	Type 7f I_P_error_correction	5.6.2.2	M	
8	Type 1h I_N_minimum delay	5.6.2.1	I	
9	Type 2h I_N_normal delay	5.6.2.1	I	
10	Type 3h I_P_error_detection	5.6.2.1	I	
11	Type 4h I_P_error_correction	5.6.2.1	I	
12	C_S duplex	5.3.1.1	C3	
13	C_F duplex	5.3.1.1	C3	
14	Downlink CL_S only	5.7.2.1	I	

(continued)

Q.9 Service support (concluded)

Service supported				
15	Downlink CL_F + CL_S	5.7.2.1	I	
16	Downlink CL_S+ SI_N	5.7.2.1	C2	
17	Uplink CL_F only	5.7.2.2	I	
18	Uplink CL_Sonly	5.7.2.2	I	
19	Uplink without SDU	5.7.2.2	I	
20	G_F simplex	5.3.1.3	M	
21	Fast paging	11.3.3.1	O	
22	Normal paging	11.3.3.1	M	
23	Low duty cycle paging	11.3.3.1	O	
24	System identities	11.3.2	M	
25	System information	11.3.2	M	
26	Encryption	6.2.3	C1	
27	Fast setup	11.3.3.2	O	

F.2.2 Service parameters

Q.10 Service parameters

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	No. of bearers in T2f service	5.6.2.1	I			
2	No. of bearers in T3f service	5.6.2.1	I			
3	No. of bearers in T4f service	5.6.2.1	M		1 - 12	
4	Forward bearer no. in T5 service	5.6.2.2	I			
5	Special bearer no. in T5 service	5.6.2.2	I			
6	Forward bearer no. in T6 service	5.6.2.2	I			
7	Special bearer no. in T6 service	5.6.2.2	I			
8	Forward bearer no. in T7 service	5.6.2.2	M		1 - 23	
9	Special bearer no. in T7 service	5.6.2.2	M		1	

F.2.3 Protocol parameters

Q.11 Protocol parameters

Protocol parameters supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Value**	
					Allowed	Supported
1	N200	10.2	M		10	
2	N201	10.6	I			
3	N202	11.4	M		10	
4	N203	9.2	I			

F.2.4 Messages

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	Support	Status	Support
1	TA code: CT data packet number 0	7.1.2	C3		C3	
2	TA code: CT data packet number 1	7.1.2	C3		C3	
3	TA code: identities information on connectionless bearer	7.1.2	N/A		C2	
4	TA code: identities information	7.1.2	M		M	
5	TA code: multiframe sync. and system information	7.1.2	N/A		M	
6	TA code: main escape	7.1.2	X		I	
7	TA code: MAC layer control	7.1.2	M		M	
8	TA code: paging/first PT transmission	7.1.2	M		M	
9	BA code: U-type, I_N, SI_N or I_P packet number 0	7.1.4	M		M	
10	BA code: U-type, I_P error detect or I_P packet number 1	7.1.4	M		M	
11	BA code: E-type, all C_F or CL_F, packet number 0	7.1.4	C3		C3	
12	BA code: E-type, all C_F , packet number 1	7.1.4	C3		C3	
13	BA code: E-type, not all C_F or CL_F; C_F packet number 0	7.1.4	C3		C3	
14	BA code: E-type, not all C_F ; C_F packet number 1	7.1.4	C3		C3	
15	BA code: E-type, all MAC control	7.1.4	M		M	
16	BA code: no B_field	7.1.4	N/A		O	
17	NT information	7.2.2	M		M	
18	Static system information	7.2.3.2	N/A		M	
19	Extended RF carrier information	7.2.3.3	N/A		I	
20	Fixed part capabilities	7.2.3.4	N/A		M	
21	Extended fixed part capabilities	7.2.3.5	N/A		I	
22	SARI message	7.2.3.6	N/A		O	
23	Multi-frame number	7.2.3.7	N/A		C1	
24	Q-channel escape	7.2.3.8	N/A		I	
25	Zero length page indication	7.2.4.2.3	N/A		M	
26	Short page indication	7.2.4.2.3	N/A		M	
27	Full page indication	7.2.4.2.3	N/A		O	
28	Not the last 36 bits of a long page indication	7.2.4.2.3	N/A		I	
29	The first 36 bits of a long page indication	7.2.4.2.3	N/A		I	
30	The last 36 bits of a long page indication	7.2.4.2.3	N/A		I	
31	All of a long page indication	7.2.4.2.3	N/A		I	
32	PT header extend flag	7.2.4.2.2	N/A		M	
33	PT MAC information: fill	7.2.4.3.2	N/A		M	
34	PT MAC information: blind full slot	7.2.4.3.3	N/A		M	
35	PT MAC information: other bearer	7.2.4.3.4	N/A		O	
36	PT MAC information: recommended other bearer	7.2.4.3.4	N/A		O	
37	PT MAC information: good RFP bearer	7.2.4.3.4	N/A		O	
38	PT MAC information: dummy or C/L bearer position	7.2.4.3.4	N/A		M	
39	PT MAC information: RFP identity	7.2.4.3.5	N/A		I	
40	PT MAC information: dummy or C/L bearer marker	7.2.4.3.7	N/A		C2	
41	PT MAC information: escape	7.2.4.3.6	N/A		I	
42	PT MAC information: bearer handover type 0000	7.2.4.3.8	N/A		I	
43	PT MAC information: bearer handover type 0001	7.2.4.3.8	N/A		I	
44	PT MAC information: bearer handover type 0010	7.2.4.3.8	N/A		I	
45	PT MAC information: bearer handover type 0011	7.2.4.3.8	N/A			

(continued)

Q.12 Messages in A_tail

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
46	Basic access request	7.2.5.2	I		N/A	
47	Basic bearer handover request	7.2.5.2	I		N/A	
48	Basic connection handover request	7.2.5.2	I		N/A	
49	Basic unconfirmed access request	7.2.5.2	I		N/A	
50	Basic bearer confirm	7.2.5.2	N/A		I	
51	Basic wait	7.2.5.2.3	I		I	
52	Basic bearer release	7.2.5.2	I		I	
53	Advanced access request	7.2.5.3.2	I		I	
54	Advanced bearer handover request	7.2.5.3.3	I		I	
55	Advanced connection handover request	7.2.5.3.4	I		I	
56	Advanced bearer confirm	7.2.5.3.6	I		I	
57	Advanced unconfirmed access request	7.2.5.3.5	I		I	
58	Advanced wait	7.2.5.3.7	I		I	
59	Advanced attributes-T request	7.2.5.3.8	I		I	
60	Advanced attributes-T confirm	7.2.5.3.8	I		I	
61	Advanced bandwidth-T request	7.2.5.3.9	I		I	
62	Advanced bandwidth-T confirm	7.2.5.3.9	I		I	
63	Advanced channel list	7.2.5.3.10	I		I	
64	Advanced unconfirmed dummy	7.2.5.3.11	I		I	
65	Advanced unconfirmed handover	7.2.5.3.12	I		I	
66	Advanced bearer release	7.2.5.3.13	I		I	
67	Transmit on the specified slot/frequency test message	7.2.5.4.2	N/A		M	
68	Loopback B_field test message	7.2.5.4.3	N/A		M	
69	Defeat antenna diversity test message	7.2.5.4.4	N/A		M	
70	Force bearer handover test message	7.2.5.4.5	N/A		I	
71	Clear test mode test message	7.2.5.4.6	N/A		M	
72	Single bearer antenna switch QC message	7.2.5.5	O		O	
73	All bearer antenna switch QC message	7.2.5.5	O		O	
74	Bearer handover QC message	7.2.5.5	I		I	
75	Connection handover QC message	7.2.5.5	I		I	
76	Single bearer frequency control QC message	7.2.5.5	O		O	
77	All bearer frequency control QC message	7.2.5.5	O		O	
78	C/L single transmission with 1 CL_F segment	7.2.5.6	I		I	
79	C/L single transmission with 2 CL_F segments	7.2.5.6	I		I	
80	C/L single transmission with 3 CL_F segments	7.2.5.6	I		I	
81	C/L single transmission with 4 CL_F segments	7.2.5.6	I		I	
82	1st C/L half slot transmission with 1 CL_F segment	7.2.5.6	I		I	
83	1st C/L full slot transmission with 4 CL_F segment	7.2.5.6	I		I	
84	2nd C/L transmission with 1 CL_F segment	7.2.5.6	I		I	
85	2nd C/L transmission with 2 CL_F segment	7.2.5.6	I		I	
86	2nd C/L transmission with 3 CL_F segment	7.2.5.6	I		I	
87	2nd C/L transmission with 4 CL_F segment	7.2.5.6	I		I	
88	C/L single transmission without CL_F segment	7.2.5.6	I		I	
89	1st C/L transmission without CL_F segment (CL_S service)	7.2.5.6	I		I	
90	Change dummy position	7.2.5.6	O		O	
91	Extended system information	7.2.5.6	O		O	

(continued)

Q.12 Messages in A_tail (concluded)

Item no.	Name	Ref.	Sending		Receipt	
			Status	**Support**	**Status**	**Support**
92	Start encryption: request	7.2.5.7	C1		N/A	
93	Start encryption: confirm	7.2.5.7	N/A		C1	
94	Start encryption: grant	7.2.5.7	C1		N/A	
95	Stop encryption: request	7.2.5.7	C1		N/A	
96	Stop encryption: confirm	7.2.5.7	N/A		C1	
97	Stop encryption: grant	7.2.5.7	C1		N/A	
98	First transmission for B_field setup	7.2.5.8	M		M	
99	M-channel escape	7.2.5.9	I		I	

Q.13 Messages in B_field

Supported messages						
Item no.	**Name**	**Ref.**	**Sending**		**Receipt**	
			Status	Support	Status	Support
1	B_field access request	7.3.3.2	M		O	
2	B_field bearer handover request	7.3.3.2	I		I	
3	B_field connection handover request	7.3.3.2	I		I	
4	B_field unconfirmed access request	7.3.3.2	M		O	
5	B_field bearer confirm	7.3.3.3	O		M	
6	B_field wait	7.3.3.4	M		M	
7	B_field attributes-B request	7.3.3.5	I		I	
8	B_field attributes-B confirm	7.3.3.5	I		I	
9	B_field bandwidth-B request	7.3.3.6	M		M	
10	B_field bandwidth-B confirm	7.3.3.6	M		M	
11	B_field channel list	7.3.3.7	M		M	
12	B_field unconfirmed dummy	7.3.3.8	I		I	
13	B_field unconfirmed handover	7.3.3.9	I		I	
14	B_field bearer release	7.3.3.10	M		M	
15	B_field null message	7.3.4	M		M	
16	B_field single bearer antenna switch QC message	7.3.5.2	O		O	
17	B_field all bearer antenna switch QC message	7.3.5.2	O		O	
18	B_field bearer handover QC message	7.3.5.2	I		I	
19	B_field connection handover QC message	7.3.5.2	I		I	
20	B_field single bearer frequency control QC message	7.3.5.2	O		O	
21	B_field all bearer frequency control QC message	7.3.5.2	O		O	
22	B-field reset request	7.3.5.3	M		M	
23	B_field reset confirm	7.3.5.3	M		M	
24	MAC-MOD2-ACK message	7.3.5.4	M		M	
25	B-field request for QT field	7.3.6.2	O		N/A	
26	B-field QT field response	7.3.6.3	N/A		O	
27	B_field TARI message	7.3.6.4	N/A		O	
28	B_field G_F data	7.3.7	M		M	
29	B_field M-channel escape	7.3.8	O		O	

F.2.5 Message parameters

Q.14 Parameters of static system information message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Normal-reverse	7.2.3.2.2	M		0 - 1	
2	Slot number	7.2.3.2.3	M		0 - 11	
3	Start position	7.2.3.2.4	M		0	
4	Escape	7.2.3.2.5	M		0 - 1	
5	Number of transceivers	7.2.3.2.6	M		0 - 3	
6	Extended RF carrier flag	7.2.3.2.7	M		0	
7	RF carriers available	7.2.3.2.8	M		1 - 1 023	
8	Spare	7.2.3.2.11	M		0	
9	Carrier number	7.2.3.2.10	M		0 - 9	
10	Spare	7.2.3.2.11	M		0	
11	Primary receiver scan number	7.2.3.2.12	M		0 - 9	

Q.15 Parameters of FP capabilities message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Extended FP information	7.2.3.4.2	M		0, 1	
2	Full slot	7.2.3.4.2	M		1	
3	Frequency control	7.2.3.4.2	M		0, 1	
4	Page repetition	7.2.3.4.2	M		0, 1	
5	C/O setup on dummy allowed	7.2.3.4.2	M		C2 = 0	
6	C/L uplink	7.2.3.4.2	M		0, 1	
7	C/L downlink	7.2.3.4.2	M		C2 = 1	
8	Basic A_field setup	7.2.3.4.2	M		0, 1	
9	Advanced A_field setup	7.2.3.4.2	M		0, 1	
10	B_field setup	7.2.3.4.2	M		1	
11	C_F messages	7.2.3.4.2	C3		0, 1	
12	I_N_minimum_delay	7.2.3.4.2	M		0, 1	
13	I_N_normal_delay	7.2.3.4.2	M		0, 1	
14	I_P_error_detection	7.2.3.4.2	M		0, 1	
15	I_P_error_correction	7.2.3.4.2	M		1	
16	Multibearer connection	7.2.3.4.2	M		0, 1	

The higher layer information field of the FP capabilities message is out of scope for mobility class 1 applications of this profile.

Q.20 Parameters of B_field access request message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	IBCN	7.3.3.2	M		0, 3	
2	Exchanged connection number	8.1.1	M		0 - 15	

(continued)

Q.20 Parameters of B_field access request message (concluded)

3	**Logical bearer number**	**10.2.4**	**M**		**0 - 15**	
4	Connection type	7.3.3.2	M		0 - 3	
5	Service type	7.3.3.2	M		3	
6	Maximum life	7.3.3.2	M		0 - 7	
7	Slot type	7.3.3.2	M		0	

Q.24 Parameters of B_field bearer confirm message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Exchanged connection number	8.1.1	M		0 - 15	
2	Logical bearer number	10.2.4	M		0 - 15	
3	Connection type	7.3.3.3	M		0 - 3	
4	Service type	7.3.3.3	M		3	
5	Maximum life	7.3.3.3	M		0 - 7	
6	Slot type	7.3.3.3	M		0	

Q.27 Parameters of B_field Bandwidth-B Request Message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Minimum number of uplink bearers	7.3.3.6	M		1 - 12	
2	Target number of uplink bearers	7.3.3.6	M		1 - 12	
3	Minimum number of downlink bearers	7.3.3.6	M		1 - 12	
4	Target number of downlink bearers	7.3.3.6	M		1 - 12	

Q.28 Parameters of B_field bandwidth-B confirm message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Minimum number of uplink bearers	7.3.3.6	M		1 - 12	
2	Target number of uplink bearers	7.3.3.6	M		1 - 12	
3	Minimum number of downlink bearers	7.3.3.6	M		1 - 12	
4	Target number of downlink bearers	7.3.3.6	M		1 - 12	

Q.29 Parameters of B_field channel list message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	First message type	7.3.3.7	M		0 - 7	
2	Second message type	7.3.3.7	M		0 - 7	
3	Third message type	7.3.3.7	M		0 - 7	

Q.32 Parameters of B_field bearer release message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Reason for release	7.3.3.10	M		0-3 and 6-13	

Q.33 Parameters of B_field null message

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	C_F data indication	7.3.4	C3		0-4	

F.2.6 Functions implemented

Q.34 Function implemented

Supported functions				
Item no.	**Function name**	**Ref.**	**Status**	**Support**
1	B_field data scrambling	6.2.4	M	
2	B_field data unscrambling	6.2.4	M	
3	R-CRC generation	6.2.5.2	M	
4	R-CRC checking	6.2.5.2	M	
5	X-CRC generation	6.2.5.4	M	
6	X-CRC checking	6.2.5.4	O	
7	Z-CRC generation	ETS 300 175-2 [2] subclause 4.8	I	
8	Z-CRC checking	ETS 300 175-2 [2] subclause 4.8	I	
9	D-MAP D80 field mapping	6.2.1.1	I	
10	D-MAP D32 field mapping	6.2.1.1	M	
11	D-MAP D08 field mapping	6.2.1.1	I	
12	D-MAP D00 field mapping	6.2.1.1	O	
13	A-MAP A_field mapping	6.2.1.2	M	
14	E/U-MUX E80 type multiplex	6.2.2.2	I	
15	E/U-MUX E32 type multiplex	6.2.2.2	M	
16	E/U-MUX E08 type multiplex	6.2.2.2	I	
17	E/U-MUX U80a type multiplex	6.2.2.2	I	
18	E/U-MUX U80b type multiplex	6.2.2.2	I	
19	E/U-MUX U32a type multiplex	6.2.2.2	C2	
20	E/U-MUX U32b type multiplex	6.2.2.2	M	
21	E/U-MUX U08a type multiplex	6.2.2.2	I	
22	E/U-MUX U08b type multiplex	6.2.2.2	I	
23	C-MUX B_field full slot mode 0 multiplex	6.2.2.3.1	M	
24	C-MUX B_field full slot mode 1 multiplex	6.2.2.3.1	C3	
25	C-MUX B_field full slot mode 2 multiplex	6.2.2.3.1	C3	
26	C-MUX B_field full slot mode 3 multiplex	6.2.2.3.1	C3	
27	C-MUX B_field full slot mode 4 multiplex	6.2.2.3.1	M	
28	C-MUX B_field half slot mode 0 multiplex	6.2.2.3.2	I	
29	C-MUX B_field half slot mode 1 multiplex	6.2.2.3.2	I	
30	T-MUX Tail multiplex	6.2.2.1.2	M	
31	Frequency correction function	11.5.2.2	O	

F.2.7 Timer support

Q.35 Timer support

Timer supported						
Item no.	**Name**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	T200	10.2	M		3 seconds	
2	T201	11.5	M		5 seconds	
3	T202	10.6	I		3 seconds	
4	T203	10.6	I		16 frames	
5	T204	9.1	M		6 multiframes	
6	T205	9.1	M		10 seconds	
7	T206	11.2	I		10 frames	
8	T207	11.3	M		5 seconds	
9	T208	11.3	M		20 seconds	
10	T209	11.4	M		30 seconds	
11	T210	11.4	M		2 seconds	
12	T211	10.3	M		3 seconds	
13	T212	10.5	M		20 frames	
14	T213	10.7	M		20 frames	
15	T214	9.2	I		20 frames	
16	T215	9.2	I		6 multiframes	

F.2.8 Procedure support

Q.36 Procedure support

Procedures supported				
Item no.	**Name of procedure**	**Ref.**	**Status**	**Support**
1	Downlink connectionless procedure	9.1.2	C2	
2	Downlink broadcast procedure	9.1.1	M	
3	Uplink connectionless procedure	9.2.3	I	
4	Request for specific Q information procedure	9.3.1	O	
5	Request for a new dummy procedure	9.3.2	O	
6	Basic setup procedure for single bearer basic connection of known service	10.2.4.2	I	
7	Normal setup procedure for single bearer advanced connection of known service	10.2.4.2	M	
8	Fast setup procedure for single bearer advanced connection of known service	10.2.4.2	O	
9	Normal setup procedure for multi-bearer symmetric connection	10.2.4.3.1	M	
10	Fast setup procedure for multi-bearer symmetric connection	10.2.4.3.1	O	
11	Normal setup procedure for full asymmetric downlink connection	10.2.4.3.3	M	
12	Fast setup procedure for full asymmetric downlink connection	10.2.4.3.3	O	
13	Normal setup procedure for full asymmetric uplink connection	10.2.4.3.2	M	
14	Fast setup procedure for full asymmetric uplink connection	10.2.4.3.2	O	
15	Setup procedure for connection with unknown service	10.2.4.3	I	
16	Connection modification procedure	10.3	M	
17	Connection release procedure	10.4	M	
18	Basic single bearer setup procedure	10.5.1.1	I	
19	A_field advanced single bearer setup procedure	10.5.1.2	I	
20	B_field single bearer normal setup procedure	10.5.1.3.1	M	
21	B_field single bearer fast setup procedure	10.5.1.3.2	O	
22	Double simplex bearer indirect setup procedure	10.5.1.4	M	
23	Double simplex bearer direct setup procedure	10.5.1.4	M	

(continued)

Q.36 Procedure support (concluded)

Procedures supported				
24	Duplex bearer handover procedure	10.6.2	I	
25	Double simplex bearer handover procedure	10.6.3	I	
26	Unacknowledged bearer release procedure	10.7.2.1	M	
27	Acknowledged bearer release procedure	10.7.2.2	I	
28	Fast bearer release procedure	10.7.2.3	I	
29	Q1 and Q2 bits setting procedure for C-channel data	10.8.1	C3	
30	Q1 and Q2 bits setting procedure for I_N and I_P_error_detection data	10.8.1.3	I	
31	BCK and Q2 bits setting for I_P_error_correction data in symmetric connection	10.8.2.4.1	M	
32	BCK and ACK bit setting for I_P_error_correction data in asymmetric connection	10.8.2.4.1	M	
33	Bearer replacement procedure	10.8.2.5.1	M	
34	MAC message jump procedure	10.8.2.5.2	M	
35	Unilateral jump procedure	10.8.2.5.3	M	
36	Idle_locked state entering procedure	11.3.2	N/A	
37	Idle_locked state maintaining procedure	11.3.3	N/A	
38	Duplex bearer channel selection procedure	11.4.1	M	
39	Double simplex bearer channel selection procedure	11.4.1	M	
40	Simplex bearer channel selection procedure	11.4.1	M	
41	Uplink connectionless channel selection	9.2.2	I	
42	RFPI handshaking procedure	11.5.1	M	
43	PT frequency correction procedure	11.5.2.2	O	
44	MAC layer test message procedure	12.2	M	
45	Receiver scanning procedure	11.9	M	
46	PP paging procedure	9.1.3.2	M	
47	Channel list procedure	10.5.2	M	

Q.37 Parameters of channel selection procedure (duplex/double/simplex bearer)

Supported parameters						
Item no.	**Parameter**	**Ref.**	**Status**	**Support**	**Values**	
					Allowed	**Supported**
1	Lowest boundary of channel list	11.4.1	M		< = - 93 dBm	
2	Band resolution	11.4.1	M		6 dB	
3	RSSI variation between checking	11.4.1	M		< = 12 dB	

History

Document history			
September 1994	Public Enquiry	PE 70:	1994-09-05 to 1994-12-30
October 1995	Vote	V 89:	1995-10-09 to 1995-12-01
February 1996	First Edition		

ISBN 2-7437-0486-1
Dépôt légal : Fevrier 1996